ISRA UNIVERSITY HYDERABAD
DEPARTMENT OF COMPUTER SCIENCE
FINAL YEAR PROJECT REPORT

CustomEats: Crafting Your Ideal Dish

By

Joti Bai	**(2101-BSCS003)**
Salochina	**(2101-BSCS005)**
Hina Kumari	**(2101-BSCS002)**

Supervised by

Dr. Mutee-u-Rahman

Co-Supervised by

Mir Afzal Talpur

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FACULTY OF ENGINEERING, SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

Certificate

This is to Certify that Hina Kumari, Joti Bai, and Salochina has worked under my supervision for the BS(CS) project titled CustomEats: Crafting Your Ideal Dish, as per partial requirement for the award of the degree of BS(CS) and the work is original and satisfactory.

Dr. Mutee-u-Rahman **Supervisor** **Department of Computer Science**	**Dr. Amber Baig** **Chairperson** **Department of Computer Science**

DEDICATION

We want to dedicate this project to, Dear Allah Almighty who gives us ray of hope even in the darkest times, who listens to our prayers and provides us with the best. And to our parents who work hard day and night, so that we can make something good out of life.

ACKNOWLEDGEMENT

All praise and thanksgiving are due to Allah, the Most Gracious and the Most Merciful, Alhamdulillah, for making this endeavor possible. We are incredibly appreciative of our families' unshakable faith, support and encouragement, which gave us the will and courage to finish this endeavor.

We want to express our thankfulness to Dr. Mutee-u-Rahman, our supervisor, who saw potential in this idea, kept us on the right track, and motivated us when we felt degraded about how others thought of this idea as not having any potential.

Secondly we want to express our thanks to Mir Afzal Talpur, our co-supervisor, who always gave us ideas, taught us to look at other possibilities too, and calming us even in the toughest times.

Lastly, we want to express our thanks to Asghar Ali Mughal (Lead Software Architect, IT Services), our mentor in this project, who helped us throughout this project by helping us tackle technical issues, by teaching us things we never knew before, and for always being there when we were in a problem.

When we had started this project, we had no idea how several things are done, or which tools will be required to make this idea real. We used to be so tensed about how we will be able to pull all this off. Being honest, deep down we had knew that we will never be able to make this project real.

But all thanks to Dr. Mutee-u-Rahman, Mir Afzal Talpur, and Asghar Ali Mughal. This project couldn’t have been without you.

ABSTRACT

The current food applications lack options for customers to customize their orders thus causing individuals to eat food that does not satisfy their needs, dietary restrictions, or unique tastes, making it difficult for them to order from restaurants.

Existing apps often lack features that allow individuals to personalize their meals, making it difficult for customers to get dishes that align with their preferences, dietary restrictions, or specific tastes. This limitation leads to dissatisfaction and a less enjoyable ordering experience.

CustomEats encourages users to customize their meals just the way they want. The app lets you pick ingredients, choose how spicy you want your food, and make any special requests. This means every order matches what you're looking for, making the food ordering more enjoyable and less frustrating. CustomEats sets a new benchmark in food customization industry, putting customers first and making sure everyone, no matter their needs, can find something that aligns and satisfies their needs.

CustomEats tackles the problems users face when they want to customize their meals filling a gap that other food apps have not addressed. By offering an intuitive and flexible solution, the app enhances the food ordering experience, catering to various preferences and restrictions. This project does not just fix a common issue it also raises the bar for keeping customers happy in the food business.

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INTRODUCTION

1.1 Project Background

In today’s busy world, most people are interested in having customized food that fits their own tastes, dietary preferences, cultural or religious restrictions. Even though there is a huge demand of customized food, many restaurants and food delivery apps lack in providing such customization that meets customer needs. This can result in dissatisfaction or frustration for individuals who follow strict diets, like vegetarians, or those who follow Halal or religious rules, as well as for individuals with food allergies or those who are picky eaters or cannot handle spice.

The problem becomes even bigger when it comes to travellers and outsiders who find it challenging to understand menus in other countries, concerned about the food they are ordering, will fit their dietary preferences. This highlights a clear gap between customer expectations and what current food apps provide, underlining the need for an improved solution that delivers more detailed customization options, assuring that individuals feel positive about what they are eating.

The aim of this project is to address such issues by developing a food customization app that allows customers to tailor their meals according to their own preferences. The app is intended to clearly demonstrate what ingredients are there in a food item, giving the customers information they need to make informed customization choices. The app also allows customers to save their customized orders in “favorites” option for quick reordering in future. Customers can also keep track of their order. Lastly, we aim to implement this project by collaborating with restaurants and by integrating this solution in their existing systems, which can enhance customer experience.

This chapter provides a comprehensive introduction to CustomEats, detailing the identified problem and the proposed solution. It further outlines the app's components, its scope, and the objectives the project seeks to accomplish. Finally, it concludes with a brief overview of the contents of the subsequent chapters.

1.2 Problem Statement

People frequently have a hard time finding meals that suit their eating habits and allergies, particularly when it comes to ordering from restaurants. This is even tougher in the case of vegetarians, or people practicing religious dietary laws as well as travellers navigating through foreign cuisines. The present food ordering apps do not allow customers to make the changes they want in their meals hence making them consume what they are not happy with. This project is about developing an app that allows users to tailor their orders to meet their specific dietary needs and wants, thus enabling them to order meals with confidence.

1.3 Proposed Solution

To solve this problem, we have to develop a food customization app that lets users personalize their meals, down to the smallest detail, from ingredients to spice levels. The app will provide clear information about each ingredient, offer alternatives for those with dietary restrictions, and let users save their favorite customizations for easy reordering. By collaborating with food restaurants, the app will ensure that these customized orders are accurately prepared, ultimately providing a more satisfying and personalized dining experience for users.

1.4 Components of the proposed Solution

- Recipe Customization
- Order Placement
- Real-time Order Status
- Recipe Management
- Order Processing
- User Management
- Recipe Details
- Order Fulfilment
- User-Friendly Interface

1.5 Focus & Scope

The focus of this project is to create a food customization platform empowering users to customize their meals as per their preferences and dietary needs. This application takes into account the expectations of customers and tries to help in meeting them by giving people the various options of personalization, clear ingredient details such that it would combine well with restaurant kitchens.

It is also about developing a user-friendly system, which should be easy to navigate both for the customers and chefs towards taking orders or maintaining preferences. Furthermore, this app seeks to enable travelers and all those special diets individuals have a way of confidently placing their orders without necessarily having to worry whether they will get what they are expecting.

The main objective behind this project is improving the overall dining experience thus promoting personalized services within the food industry.

1.6 Objectives

Objectives of the proposed solution are as following:

i. Through existing applications examine the customization options available as well as their limitations and conduct a survey to understand the problems customers face due to unavailability of sufficient customization features when placing an order for food.

ii. Based on the identified problem, develop and integrate the back-end system that supports customization options, allowing customers to modify ingredients, spice levels, and dietary preferences.

iii. Create transparency between customers and chefs and/or restaurants.

iv. Empower customers with the freedom to personalize their meals, ensuring every dish aligns with their dietary preferences, health choices, cultural values, and religious needs.

1.7 Organization of the Report

Chapter 1: Introduction: offers a detailed overview of CustomEats, explaining the problem it aims to address and the solution it proposes. It also describes the app's components, its scope, and the goals the project intends to achieve. The chapter concludes with a brief summary of what to expect in the following chapters.

Chapter 2, Literature Review: presents a thorough introduction to the literature review, discussing compelling reasons for the food industry to prioritize food customization. It highlights the features and functionalities that set this app apart from others that do not offer such options. The chapter wraps up with a survey aimed at gathering real-world insights.

Chapter 3, Methodology: forms the core of this project, providing a concise research overview to identify the problem, designing the screens through paper frames or prototypes, and emphasizing how the survey informed the design of the screens and app functionalities. It also details the tools and technologies used to integrate both the frontend and backend, while mentioning features that are planned for future enhancements but could not be implemented at this time.

Chapter 4, System Implementation: includes organization of the system, breakdown of system organization for a better and a clear understanding, breakdown containing descriptions and diagrams of customer, chef, and admin process flow, a separate standalone section for app’s features, containing visual representations and a brief description of functionalities like ingredients selection, cart, favorites, order status, payment options, and why login is required.

Chapter 5, Conclusion: includes restating of the purpose, summary of the key findings, interpretation of the results, limitations of this app, practical implications of this app in real world, and a detailed discussion of future enhancements.

CHAPTER 2

LITERATURE REVIEW

This chapter provides a broad introduction to literature review which discusses and provides solid reasoning why food industries should focus on providing food customization. It further outlines the features and functionalities that make this app a better choice than the ones that lack in providing such features. Finally, it winds up with a survey conducted to gain real world insights.

2.1 Introduction

Fast food chains have responded to the increasing requirement of food personalization by including customizable menus where meals are prepared according to one's dietary needs, choice, and cultural preferences. The role of technology played here is pivotal, as, with the help of smartphones and online applications, a consumer can prepare his own meal by selecting the necessary ingredients, portions, and mode of cooking to their advantage. [6]

Customizable menus can be a way to satisfy consumers' cravings and may also demonstrate consumer loyalty. Therefore, fast food chains will have to study the consumer preferences to survive the competition. Real-time preference and trend data from custom online panels like Eyes4Research managed by Rudly Raphael can be used by fast food chains for the modification of their offerings and communication/marketing strategies. [1]

The rise of personalized nutrition points out the uniqueness of individuals in terms of how they metabolize nutrients, brought about by genetic profiles and gut microbiomes. Personalized food manufacturing emphasizes the fact that food can be prepared according to unique nutritional requirements and also according to tastes, textures, and even other personal preferences through the use of traditional and advanced technologies. [2]

The rise of technology has also showed that people can trace out their food intake and diets. Food personalization brings a control over food quality and source, so they can improve their well-being both physically and mentally. [3]

Moreover, the COVID-19 pandemic has also accelerated the shift towards food personalization. Higher awareness of the health impact of food has opened avenues for healthy diets, making it possible for the food industry to provide personalized choices (CIO Applications, 2019). People have realized that diet impacts health and wellness, and this has changed their expectations to more individualized dietary choices. [5]

Therefore, personalization is the primary aspect in enhancing customer satisfaction and loyalty. The drive towards personalization by technology has helped businesses keep up with consumers' various dietary needs in an efficient manner. On the other hand, customization cannot be extreme to the point of disrupting operational efficiency since it increases costs and waiting time. Besides, the preferences of consumers can be monitored by data-driven insights, keeping fast food chains and other businesses competitive in the dynamic market. [4]

2.2 Comparison

The comparison below, of CustomEats is with restaurant app's. The comparison is made in order to show what new CustomEats offers in the food industry that existing apps failed to offer, not to make any restaurant look bad. The comparison is made with three apps, one that offers no customization, one that offers minimal customization but not guaranteed, and one that offers really good customization but only in fast food.

i. CustomEats vs. Buns & Bites

In CustomEats you can find customizable ingredient lists and customizable spice level, which no other app offers yet. If we talk about Buns & Bites, customers can only add something extra and select the kind of beverage they want. At Buns & Bites customers cannot select how spicy they want, they cannot remove lettuce, they cannot replace mayonnaise with ketchup. While Buns & Bites offers a limited customization in fast food chain, CustomEats provides fully customizable meals.

ii. CustomEats vs. Desi Meal Drop

As CustomEats is an app developed to provide customization in desi food, while on the other hand Desi Meal Drop provides both desi and fast food. But you cannot find ingredient selection lists or spice level selection at Desi Meal Drop. Desi Meal Drop only provides you a special instruction box for your orders, which is only rare that those instructions are taken into consideration, whereas CustomEats not only provides that but also the selection of ingredients of your choice.

iii. CustomEats vs. Crispy Crust

Until now, there is no other app in Pakistan that offers customization as much Crispy Crust does. At Crispy Crust you can select which type of crust you want, which flavor of pizza you want, you can also add extra toppings of your choice from veggies to meat, and of course you can select sauce and beverage of your choice. But at the end of the day, if we look at the other side of it, people cannot eat fast food all the time, they need to eat food that is not only tasty but also healthy for them. This is what CustomEats is all about, providing customization in desi food items while maintaining the nutrition of the dish.

2.3 Survey

This survey aims to find out how exactly the customers are becoming more and more inclined to change food according to their needs. We are interested in how frequently and to what extent people wish to customize their food considering their eating habits, culture, religion or medical conditions. By knowing such demands we also would be attempting to identify deficiencies within current food services and applications, more precisely the lack of ease of customization, and measure people's readiness for a proposition of how to easily and effectively customize meals.

Survey questions are divided into categories for a better understanding. In each category multiple questions are asked to cover every aspect of the challenges faced by individuals and to get suggestions from people and think about how we could do our best to help them.

2.3.1 Customer Preferences and Customization Needs

Figure 2.3.1 shows the number of percentage of people that would prefer to use an app that allows them to customize their food according to their preferences.

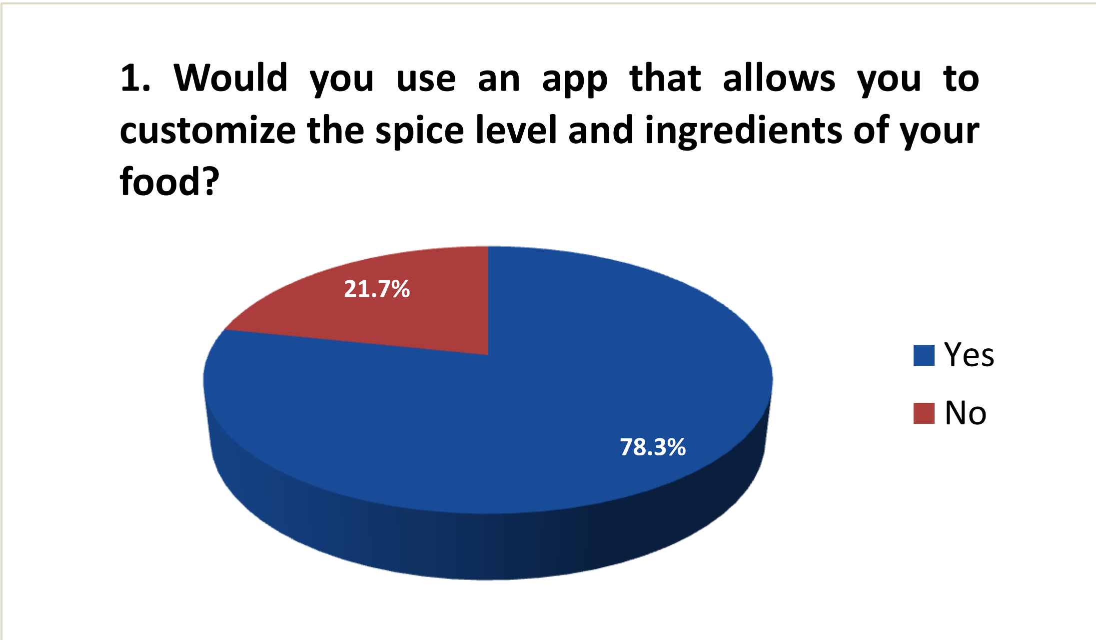


Figure 2.3.1 App Usability (customer preferences)

Figure 2.3.2 shows various responses from individuals illustrating the amount of satisfaction they have with current food apps.

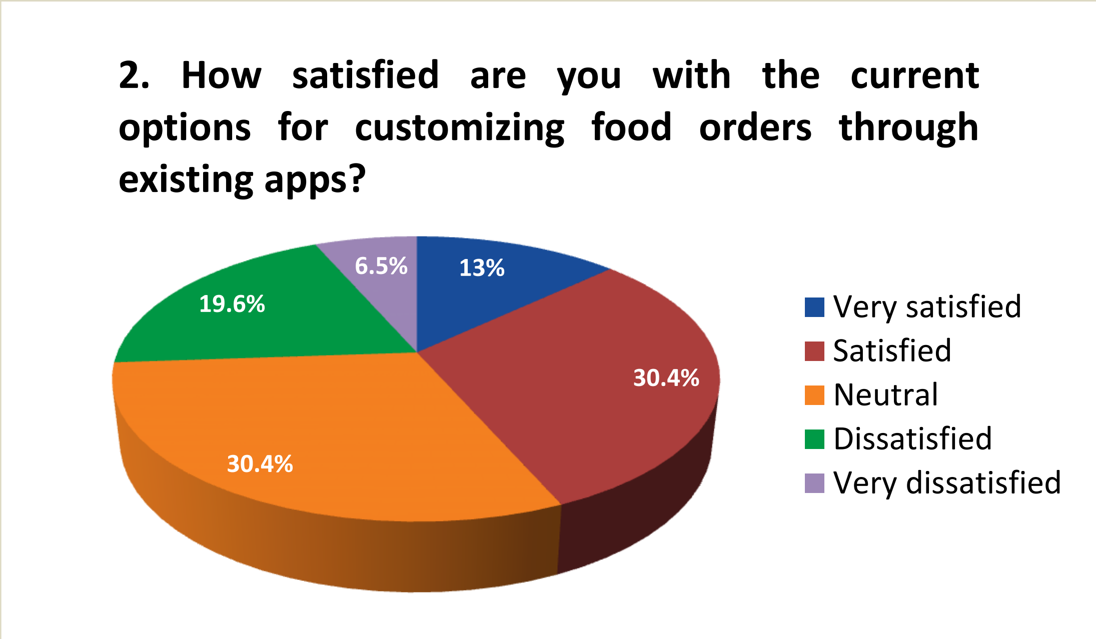


Figure 2.3.2 Customer Satisfaction with current Apps

Figure 2.3.3 shows the percentage of people who customize their food on a regular, weekly, and occasional basis, and those who never customize their food.

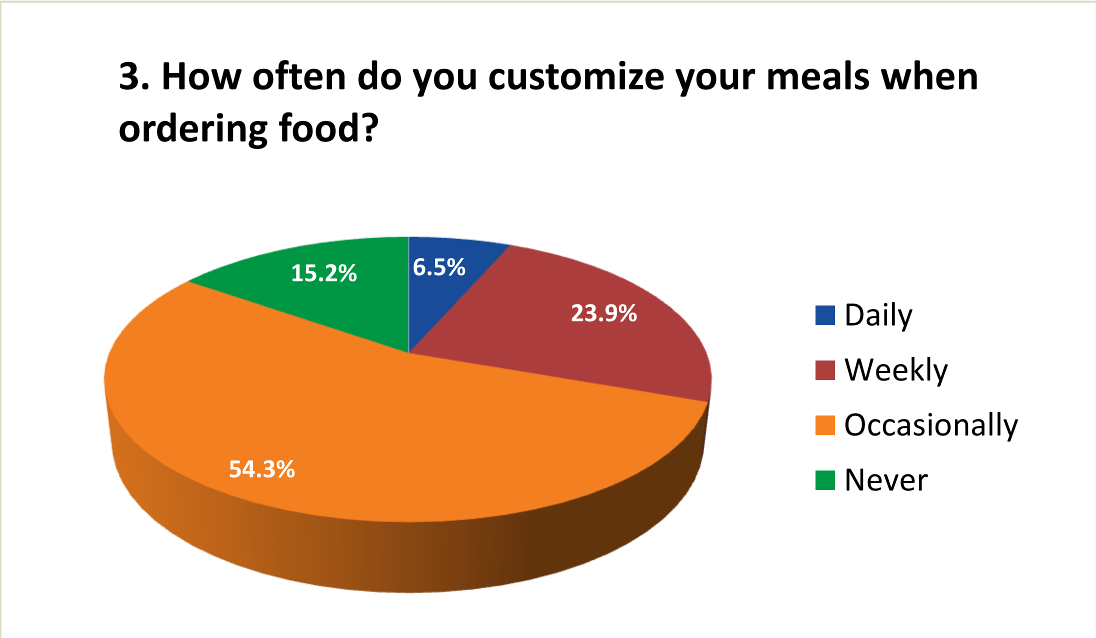


Figure 2.3.3 Food Customization on a quick basis

Figure 2.3.4 shows a huge number of individuals would be more open to trying new cuisines if there was an option to customize the dishes and selecting alternative options according to their preferences.

Figure 2.3.4 Willingness to try new Cuisines

Survey Question 05: In what ways could a food app improve your ordering experience, particularly when it comes to customization?

Figure 2.3.5 shows the responses submitted by customers, which highlights the features/functionalities that a food customization app can offer to improve customer experience.

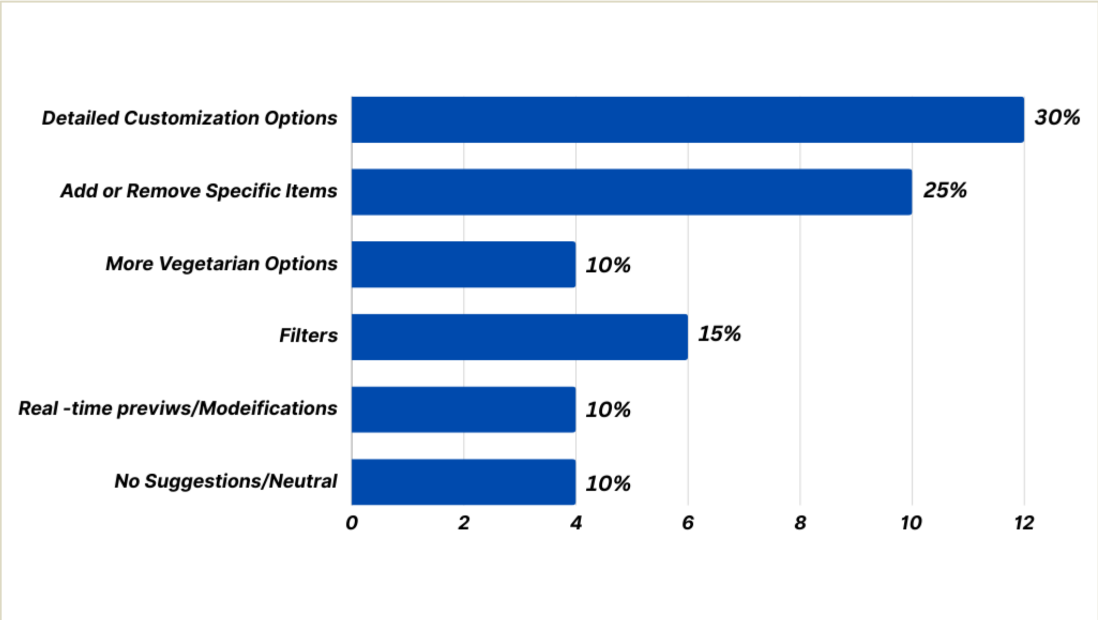


Figure 2.3.5 Features to improve Customization

2.3.2 Challenges with Dietary and Religious Requirements

Figure 2.3.6 shows that a lot of people face challenges finding food that aligns with their religious dietary requirements when ordering from restaurants.

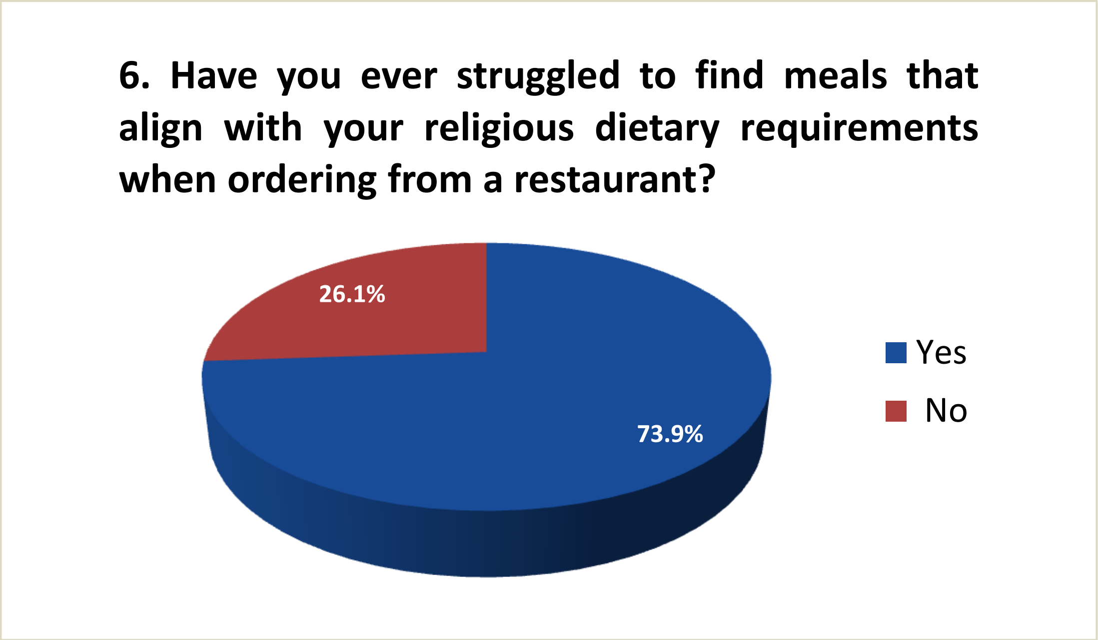


Figure 2.3.6 struggles in finding Food

Figure 2.3.7 shows that 89.1% of the individuals would feel more confident in their vegetarian choices if the app offered detailed ingredient lists that could be customized.

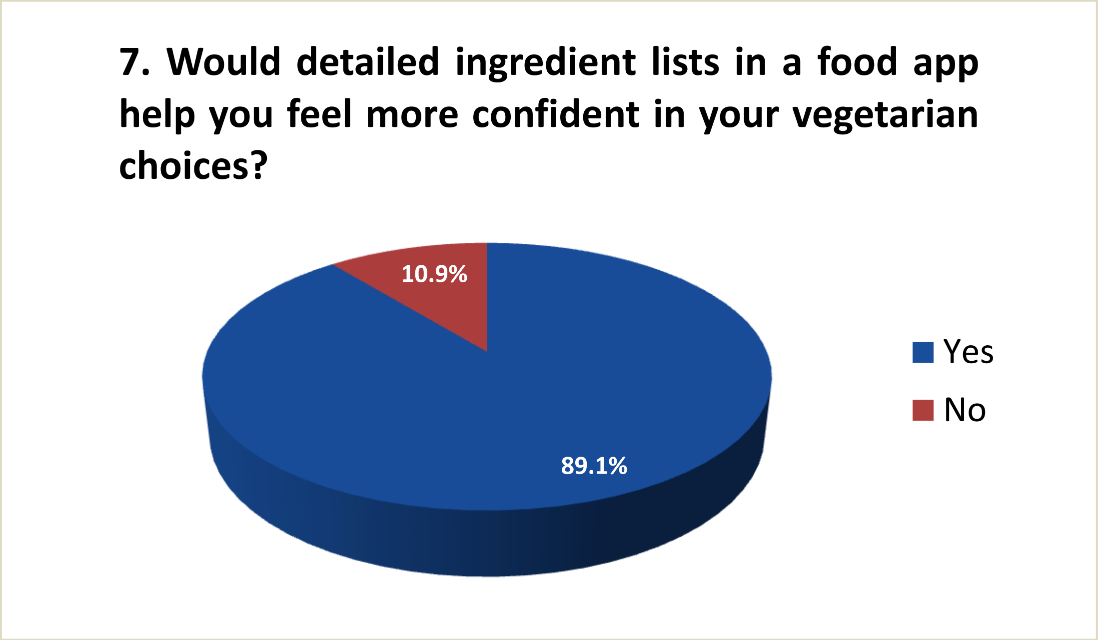


Figure 2.3.7 confident vegetarian choices

2.3.3 Past Order Experiences and Issues

Figure 2.3.8 shows that more than 50% the individuals decided not to order a dish because it included some ingredients they cannot eat, and there was no option for customization. If there was an option they probably would have ordered the dish.

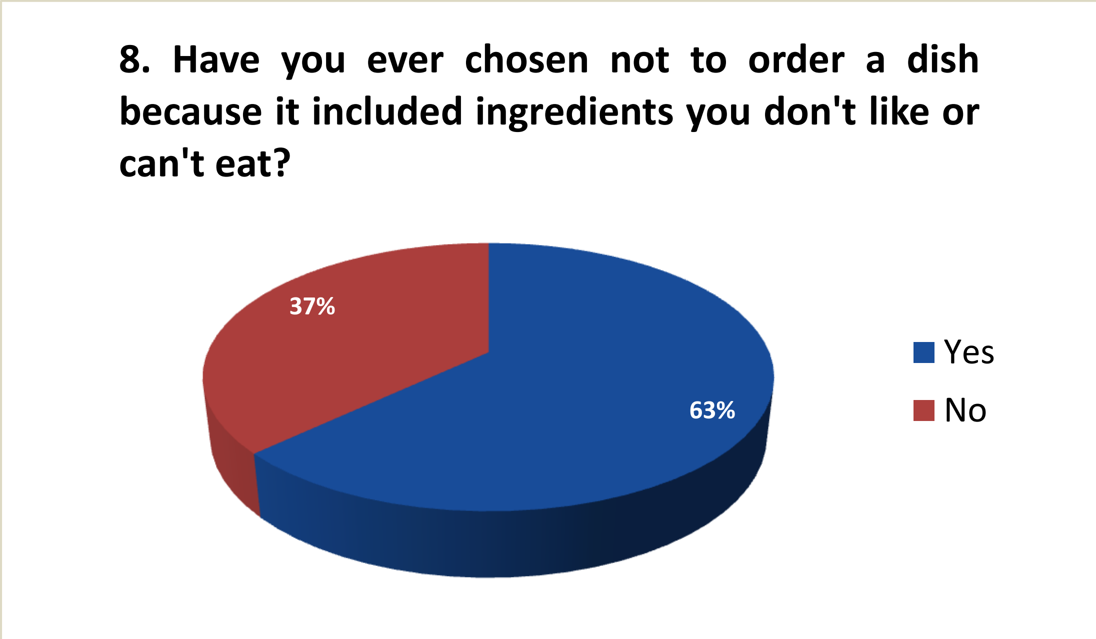


Figure 2.3.8 customer likes and dislikes

Figure 2.3.9 shows the equal amount of customer satisfaction, and equal amount of dissatisfaction caused by unclear customization options.

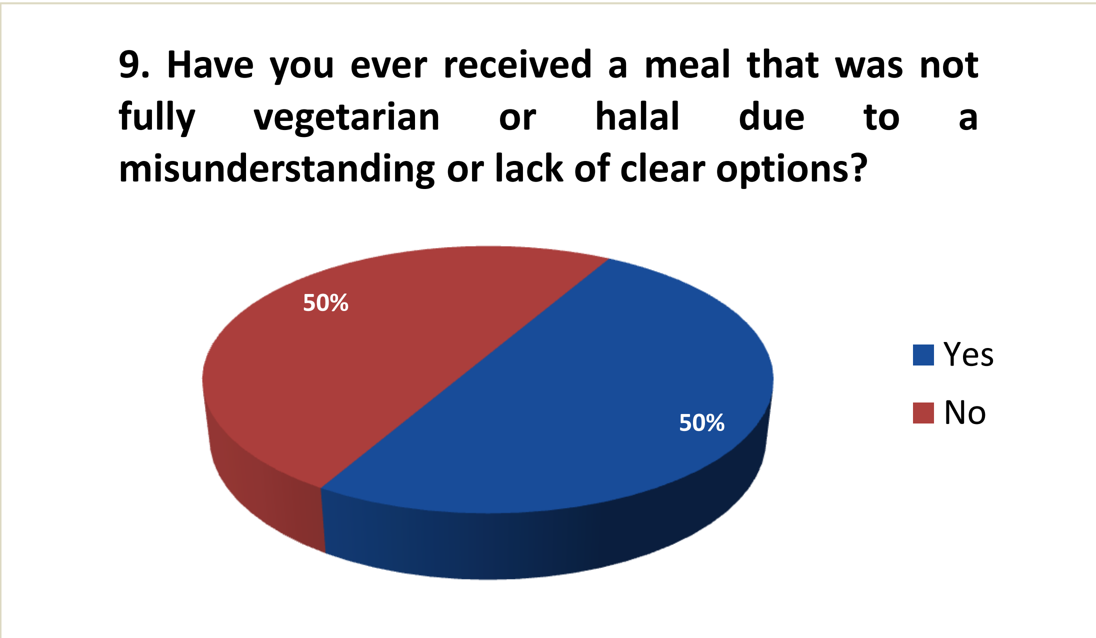


Figure 2.3.9 unclear customization options

CHAPTER 3

METHODOLOGY

This chapter serves as the backbone of this project, containing a brief research to identify the problem, designing the screens whether paper frames or prototypes, highlighting how survey contributed in the designing of screens and functionalities of the app, the tools and technologies used to integrate both frontend and backend, and including features left for future enhancements, which cannot be achieved as for the time being.

3.1 Research and Problem Identification

As research and identification being the first step of methodology, the focus was to analyse these three areas, existing apps, customer needs, and market gaps. We started by investigating current food apps to determine whether they provided customization options that satisfied customers dietary needs and religious restrictions, which is clearly demonstrated in literature review (chapter 2). Then comes, customer needs and market gaps, through the usability survey of this app we found out a huge difference between what customers expected and what restaurants offered. Clearly displayed in 2.3 (chapter 2).

3.2 Design Phase

As soon as the problem was identified, we started working on designing. The process started with making paper frames about how the app (interfaces) will look like. Initial design was modified continuously until it reached a point where it was certain that this would meet customer needs.

3.2.1 Paper Frames for Customer Interface

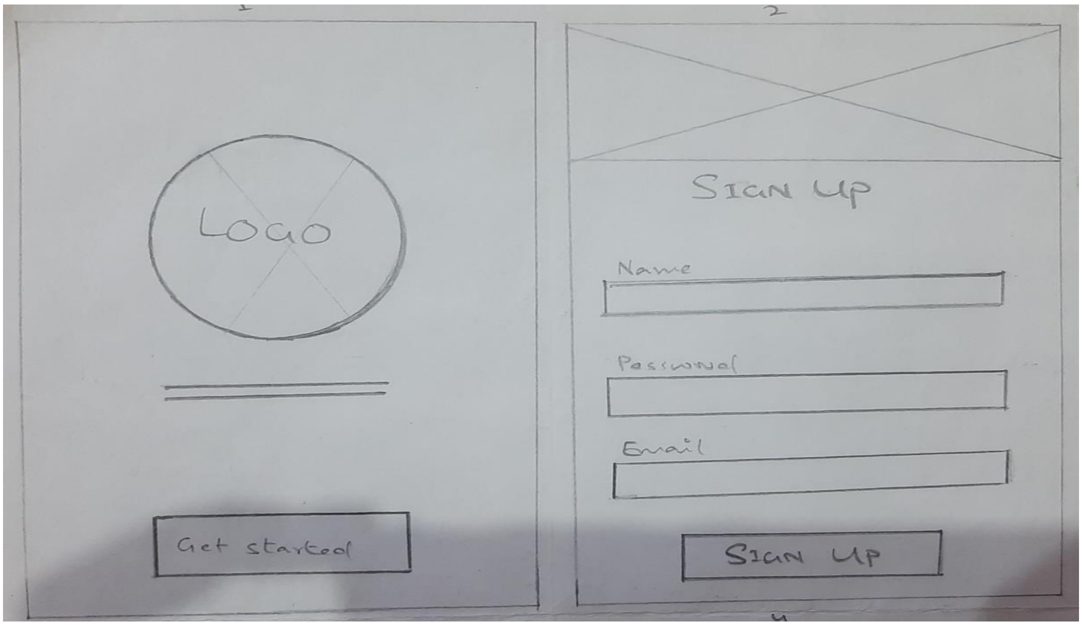


Figure 3.2.1 Splash & Login Screen

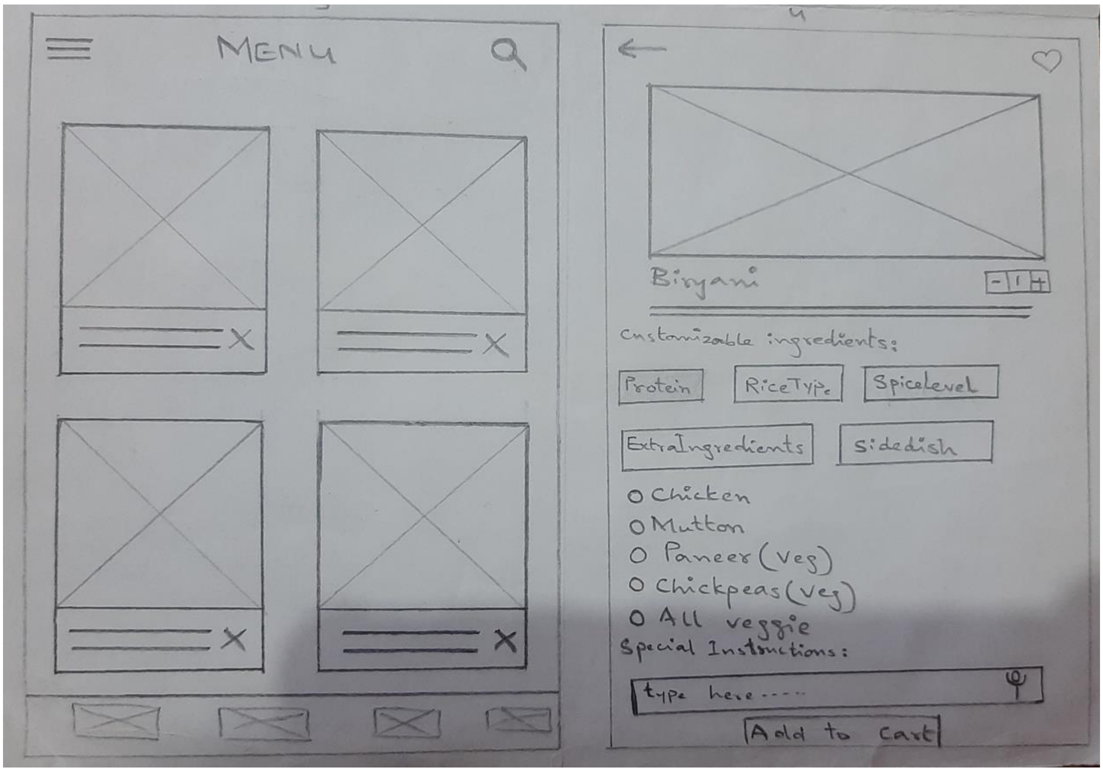


Figure 3.2.2 Homepage & Item Customization

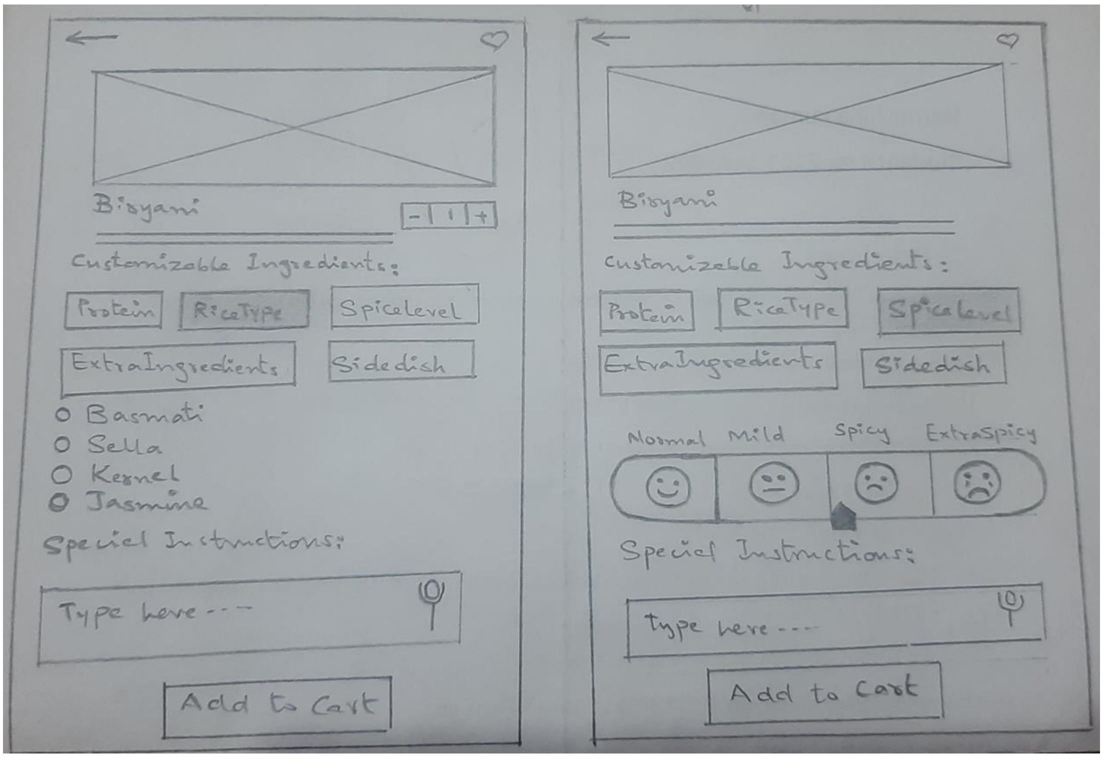


Figure 3.2.3 Continued Customization 1

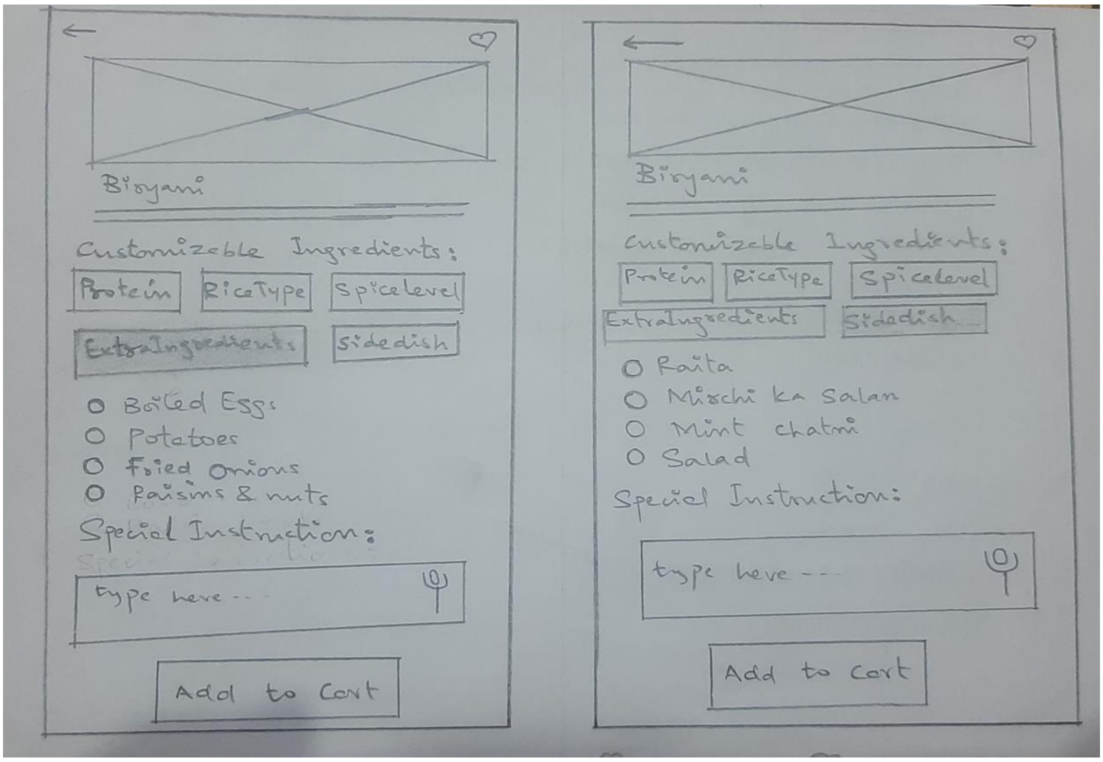


Figure 3.2.4 Continued Customization 2

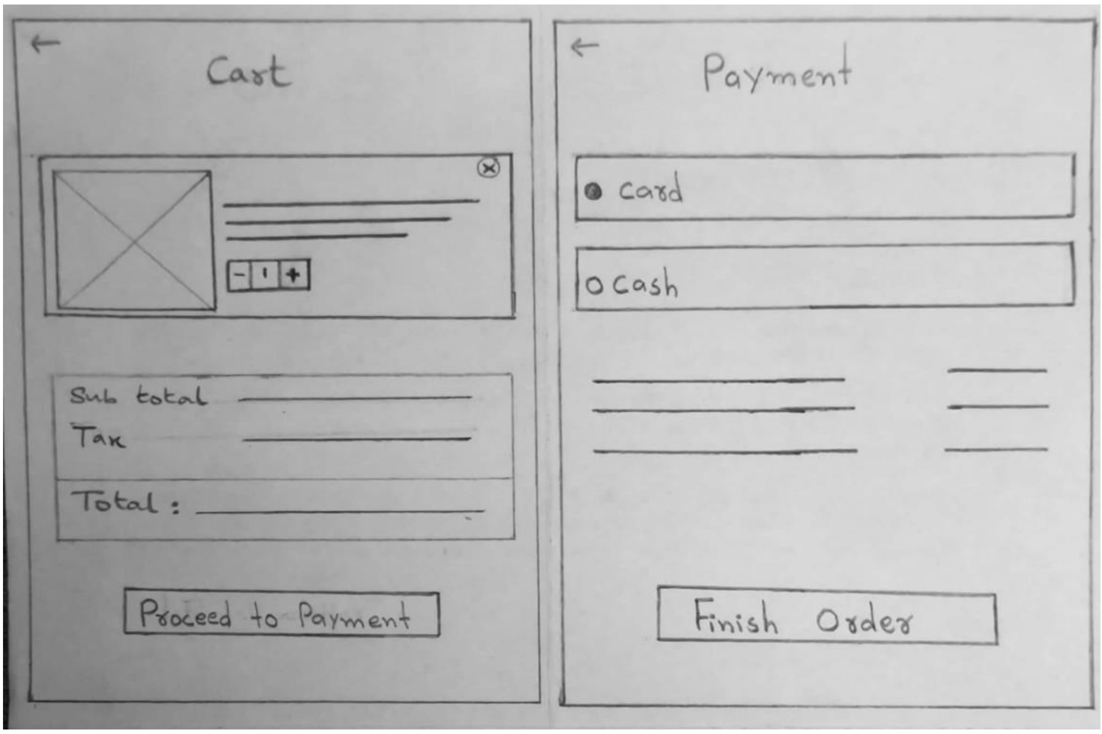


Figure 3.2.5 Cart viewing & Payment options

After being satisfied with paper frames, the process further preceded with creating prototypes for the customer interface using Figma. As it happens in the most cases, it happened with us too, it was realised that some of the elements designed in the paper frames could not be achieved, leading to necessary modifications being made in the prototypes. As shown below.

3.2.2 Prototypes for Customer Interface

When the user/customer will first open the app the splash screen would display for a few seconds and then the first screen of the app would appear that is menu page. As shown below in figure 3.2.6.

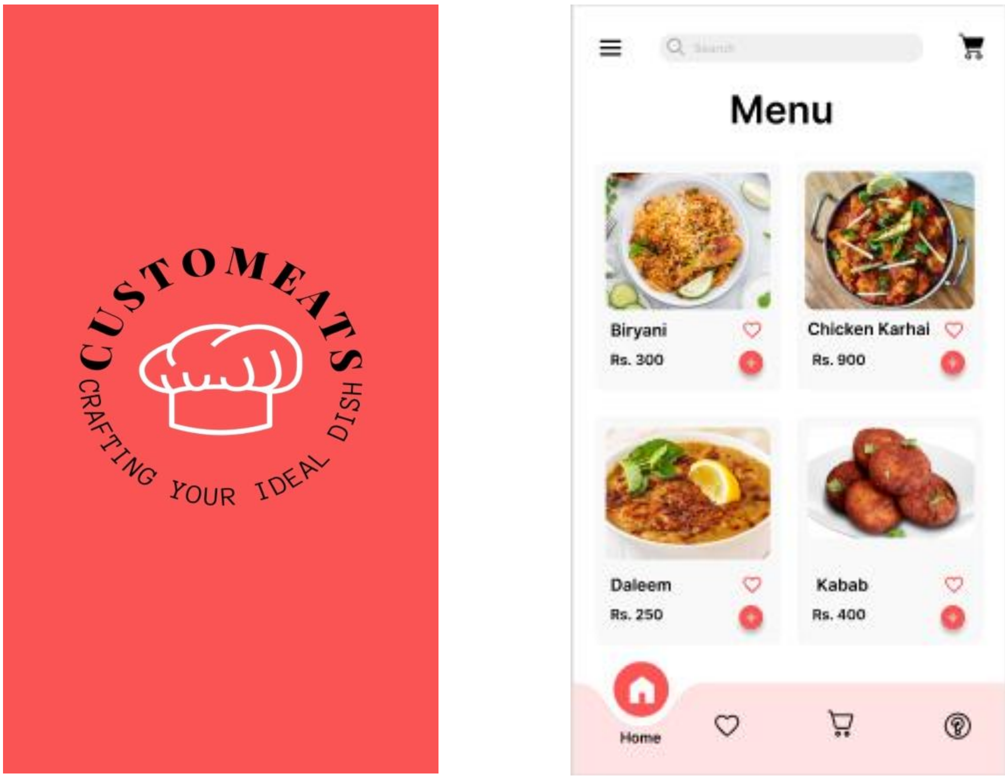


Figure 3.2.6 Splash Screen & Menu page

After the customer has selected an item from the menu he/she can customize it as preferred. The customer has to select what they like from the given list of choices. As shown below in figure 3.2.7.

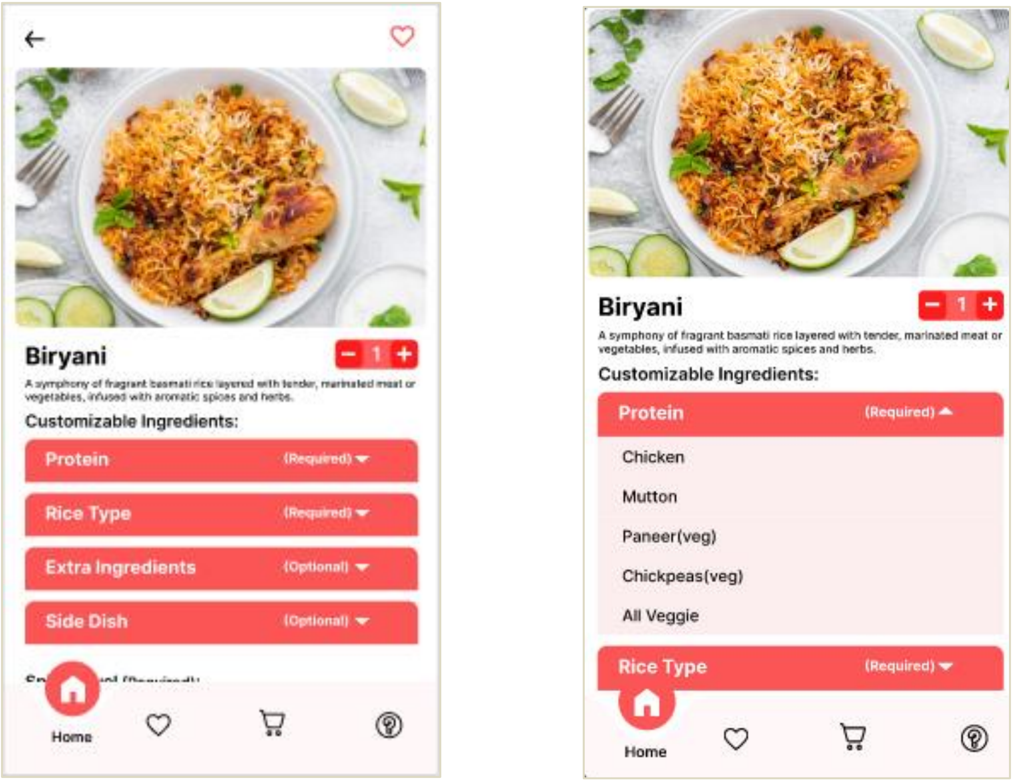
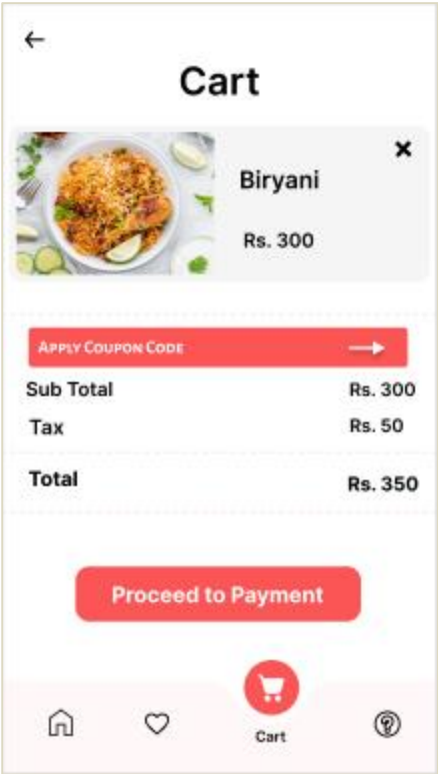


Figure 3.2.7 item Customization

Next the customer has to add the customized food item to the cart and proceed to payment in order to place their food order. As shown below in figure 3.2.8.



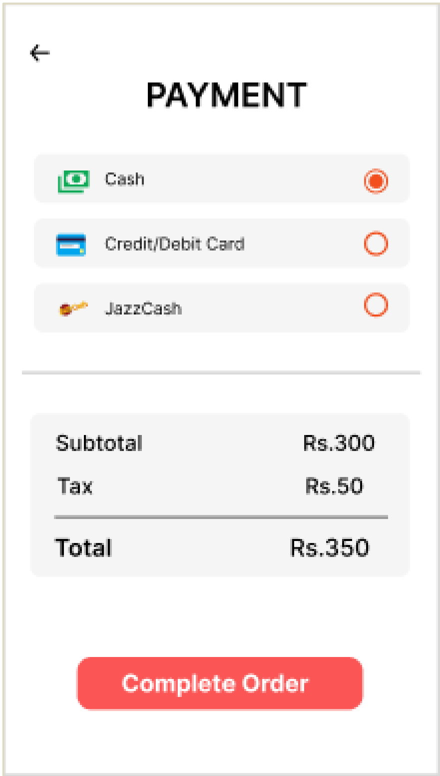


Figure 3.2.8 Cart and Payment screen

Once, the designing of customer interface was finalized, next step was to design the chef interface. The way we tried to keep things simple for customers by designing a user-friendly interface, here, the focus was to make sure that the chefs get the same. The figure 3.2.9 below shows how the orders would be displayed in the kitchen on a touch screen monitor.

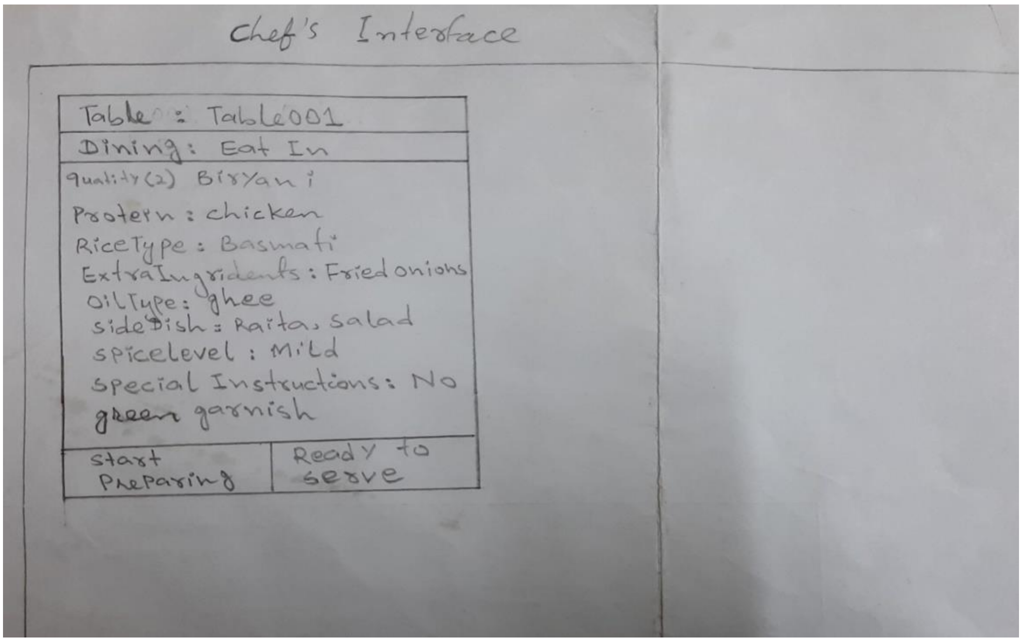


Figure 3.2.9 Kitchen display System

The design process then continued with the creation of the chef interface in Figma.

The chef has to login first in order to view the incoming orders. As shown below in figure 3.2.10.

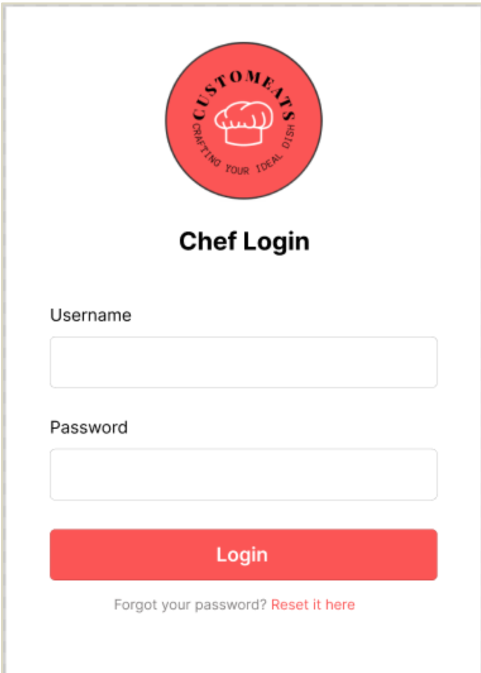


Figure 3.2.10 Chef Login

When the chef logs in to the system he/she will be navigated to where the orders are coming, if there are no orders coming he/she will be shown an empty screen. As shown below in figures 3.2.11 and 3.2.12. [7]

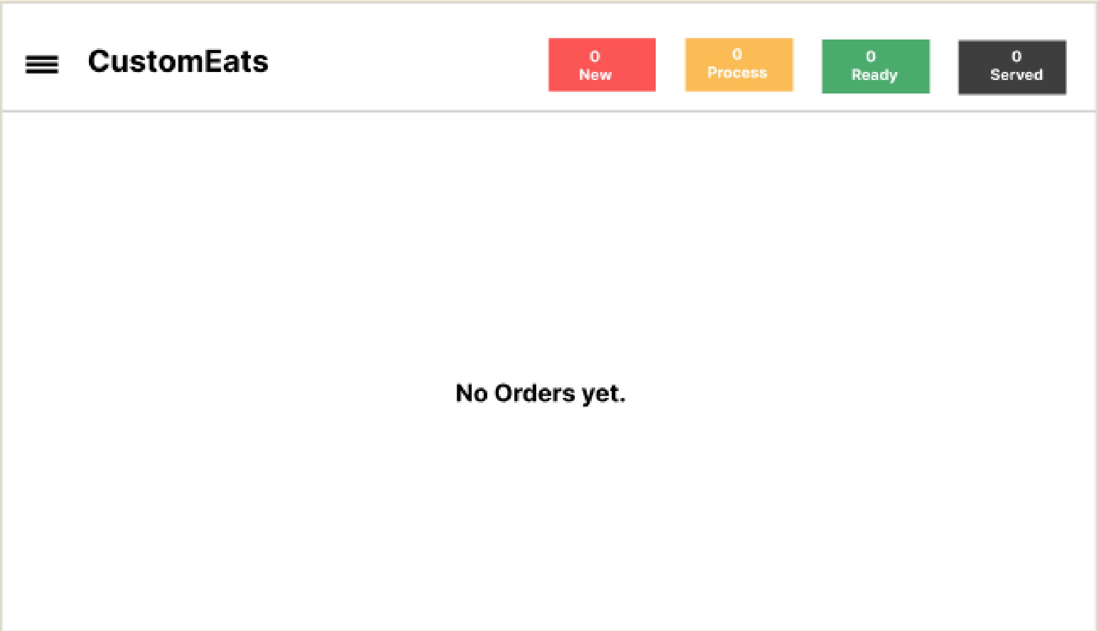


Figure 3.2.11 No Orders Yet

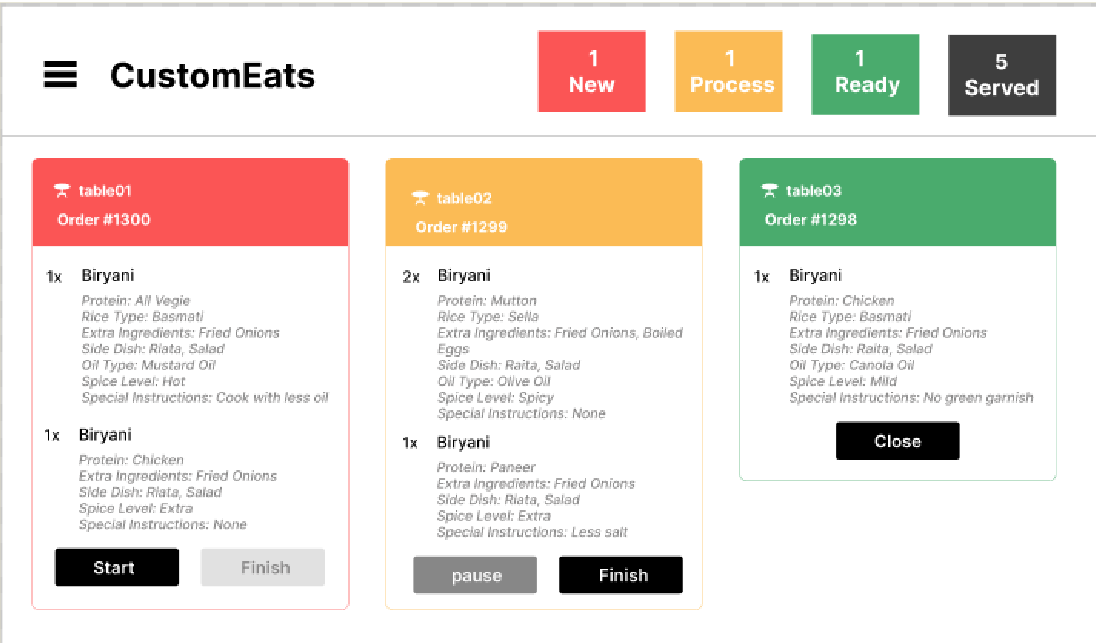


Figure 3.2.12 Incoming Orders

Lastly we have prototypes of admin Interface, where the admin can login and logout, view and manage items, orders, and staff. [8]

The admin also has to login first in order to view the dashboard. As shown below in figure 3.2.13.

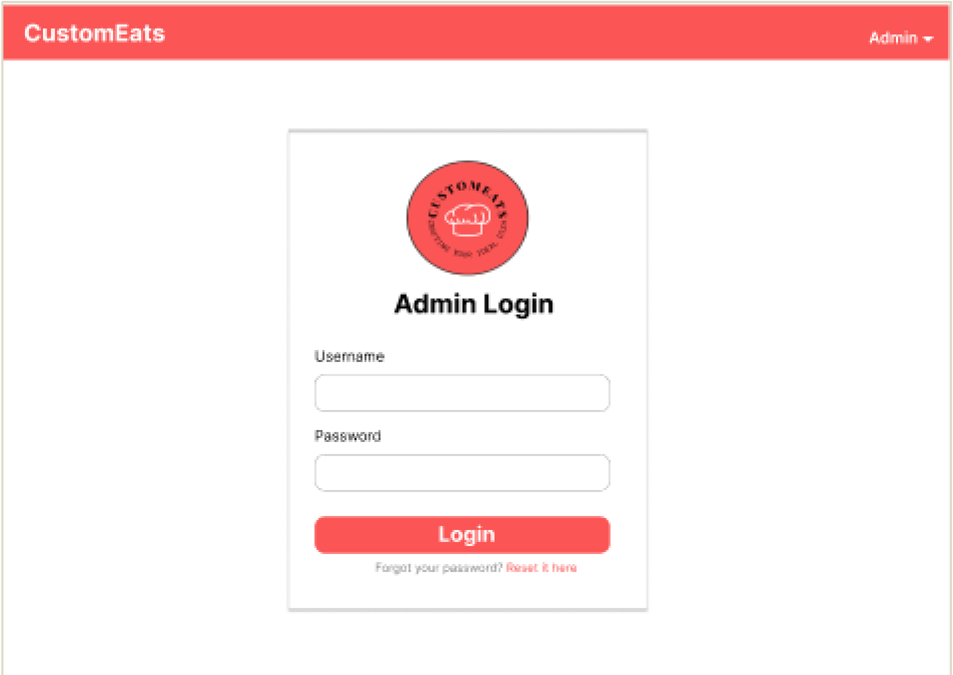


Figure 3.2.13 Admin Login

After the admin has logged in he/she can view, manage and update data as needed. The figure 3.2.14 below shows the availability and unavailability of the tables.

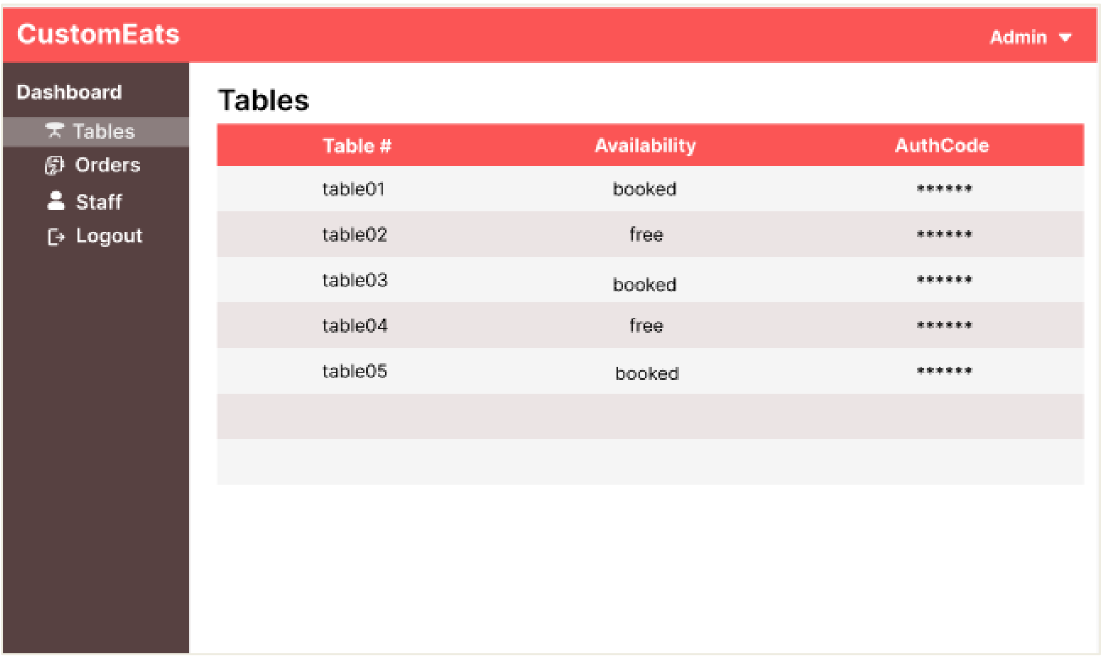


Figure 3.2.14 View Restaurant Tables

The admin can also see incoming orders but not the customizations of the dish, he/she can see what payment method the customer selected at what time and date the order was placed and if the order was delivered or cancelled. As shown below in figure 3.2.15.

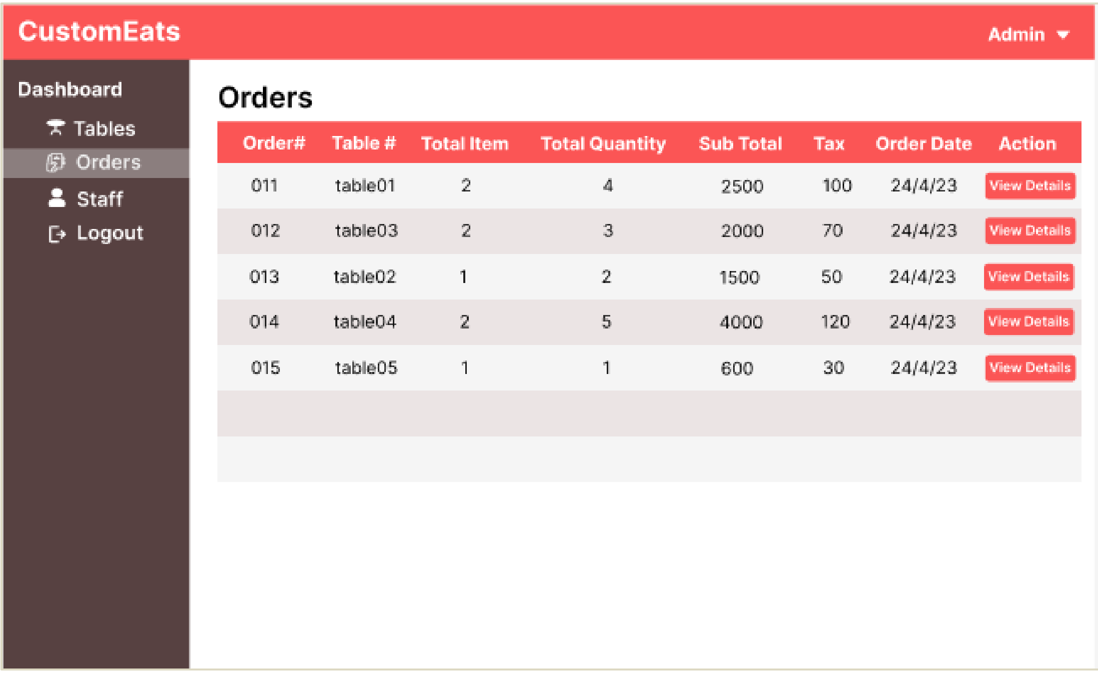


Figure 3.2.15 View Orders at Admin side

The admin can manage staff details like updating address or phone number and adding or removing information of new staff or previous staff. As shown below in figure 3.2.16.

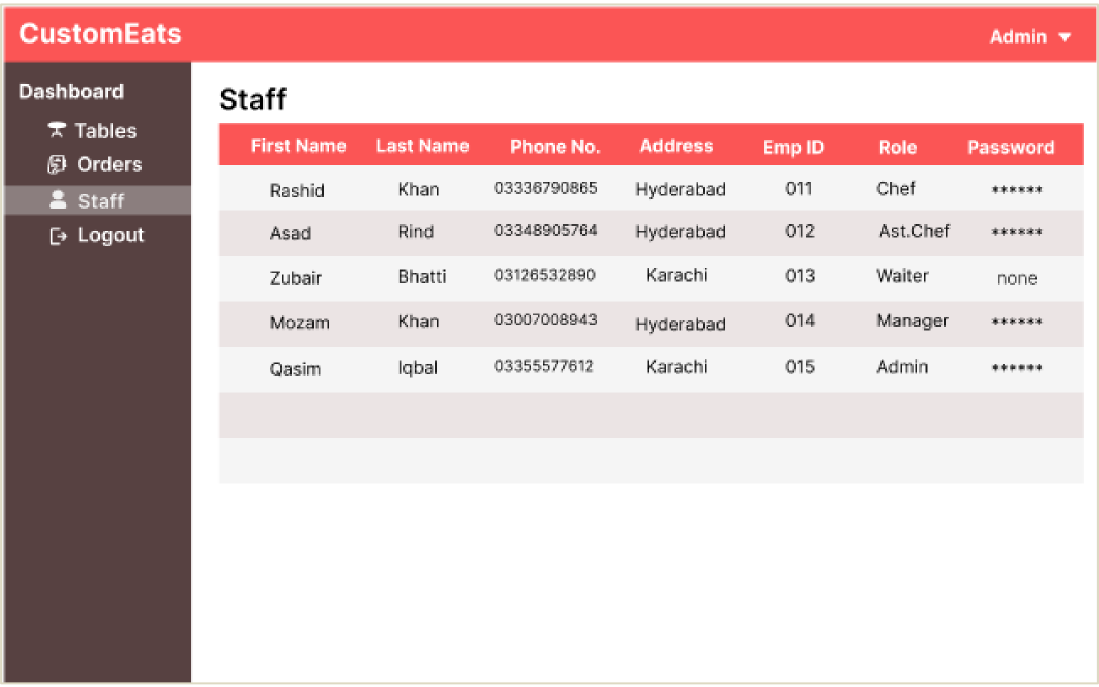


Figure 3.2.16 Staff Management

The admin can update prices, images and availability (in stock or out of stock) of food item. As shown below in figure 3.2.17.

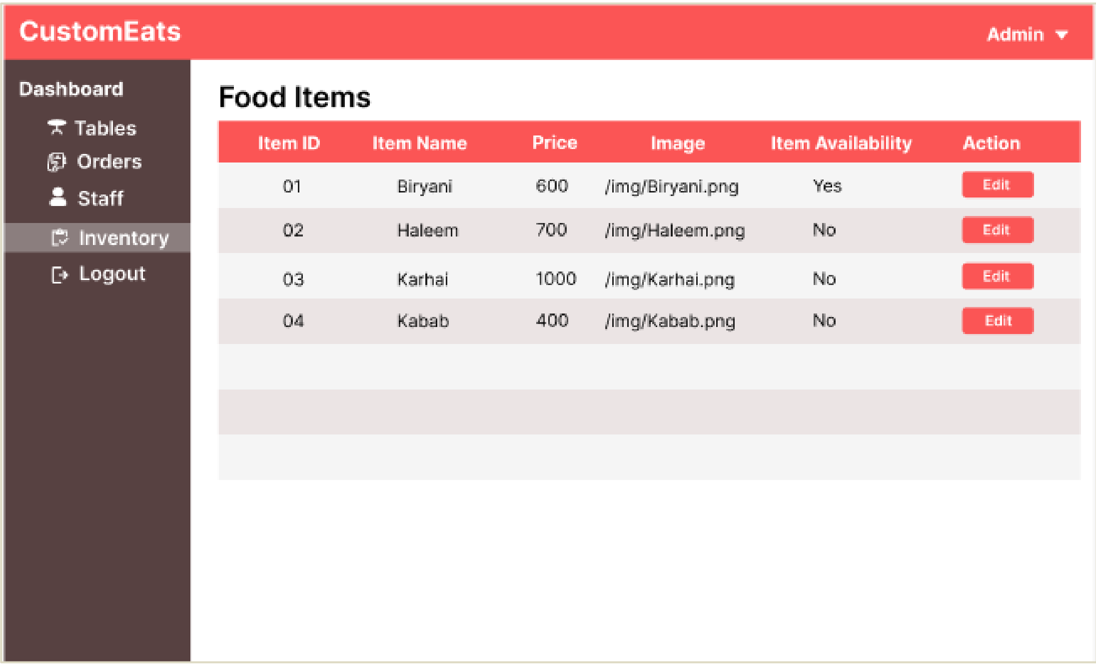


Figure 3.2.17 Inventory Management

3.3 Survey and Feedback Collection

In the third step "Survey and feedback collection", we focused on gathering real world problems faced by customers, clearly demonstrated in survey (2.3, chapter 2). The usability survey helped us pain point the needs of customers. Though the interfaces were under development and not shown to respondents when the survey was conducted, but the responses received helped in finalizing the interfaces and making sure they were up to customers' expectations.

3.4 Back-end Development and Integration

In the fourth step "Back-end development and integration", the process started with developing front-end using flutter, making sure the interfaces were completely functional and intuitive for end-users. Once the interfaces were in place, we moved on to the development of the back-end, so that the applied functionalities of the app were operable. The backend was built using Django and integrated with a MySQL database. It enabled us to manage the data of the application, which includes saving and retrieving user preferences. At this stage, the front-end was integrated with back-end to achieve smooth communication between customer and chef interface.

The tools used for Front-End and Back-End development & Integration are defined below.

i. **Android Studio**

Android Studio served as the main Integrated Development Environment (IDE) for the Front-End development of CustomEats app. Flutter SDK was used within Android Studio for front-end development using Dart language, so that the app would work on any mobile operating system (like iOS, android etc.).

ii. **API**

The API (Application Programming Interface) acted as a bridge between front-end and back-end, supporting the data exchange between CustomEats app and the server. The API was designed using JSON for lightweight data handling.

iii. **Django**

Django, a robust framework that is designed in Python, was chosen due to its simplicity in terms of backend development. Django allowed the management of tasks related to data, such as, requesting for data fetching, and sending responses back to Django.

iv. **MYSQL**

MySQL, that is famous for its relational database management handling, was chosen to for the creation, modification, and maintaining of the data needed by CustomEats. MySQL plays and important role in this app's development, by storing all the customization preferences provided by customers.

v. **WampServer**

WampServer was employed to create a local server environment, facilitating the development and testing of the Django backend on a Windows machine.

This is how the database looks. As it can be seen in figure 3.4.1, the tables have no connection, no relation between them, this type of tables are referred to as standalone tables or independent tables, these tables represent data that does not depend or relate to other tables, these tables are created to represent singular sets of information.

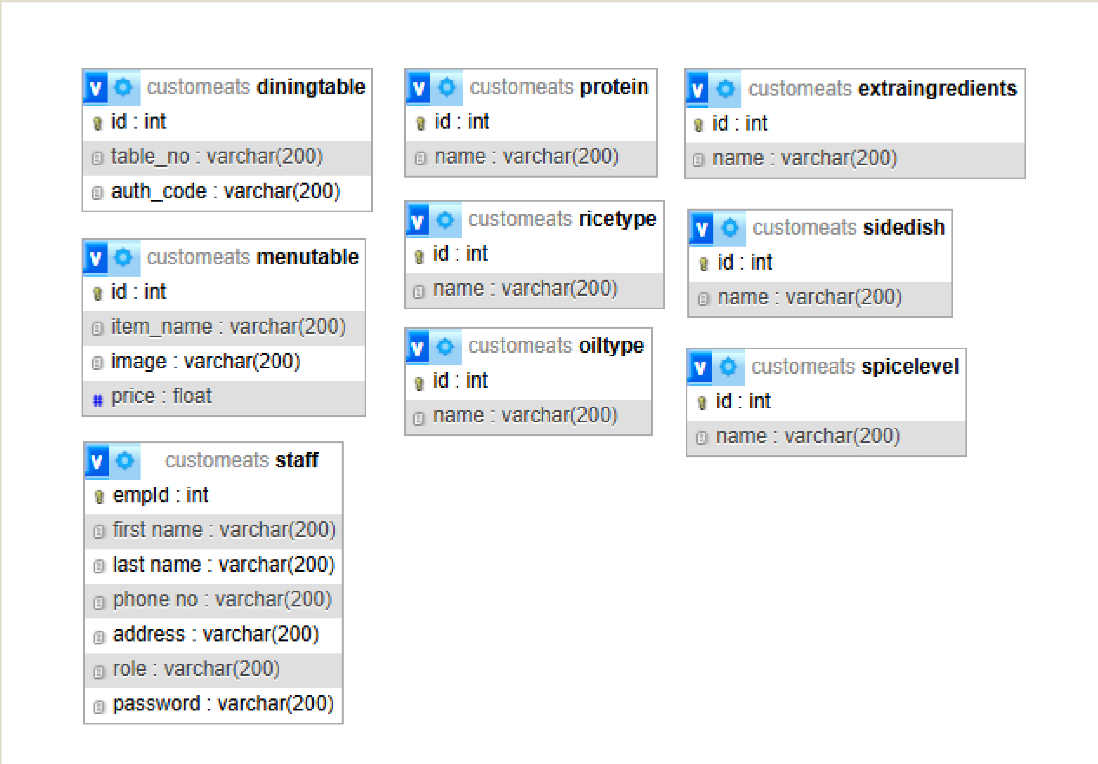


Figure 3.4.1 Database Schema

The figure 3.4.2 below shows a rough representation of the tools used in this project and their connection in between.

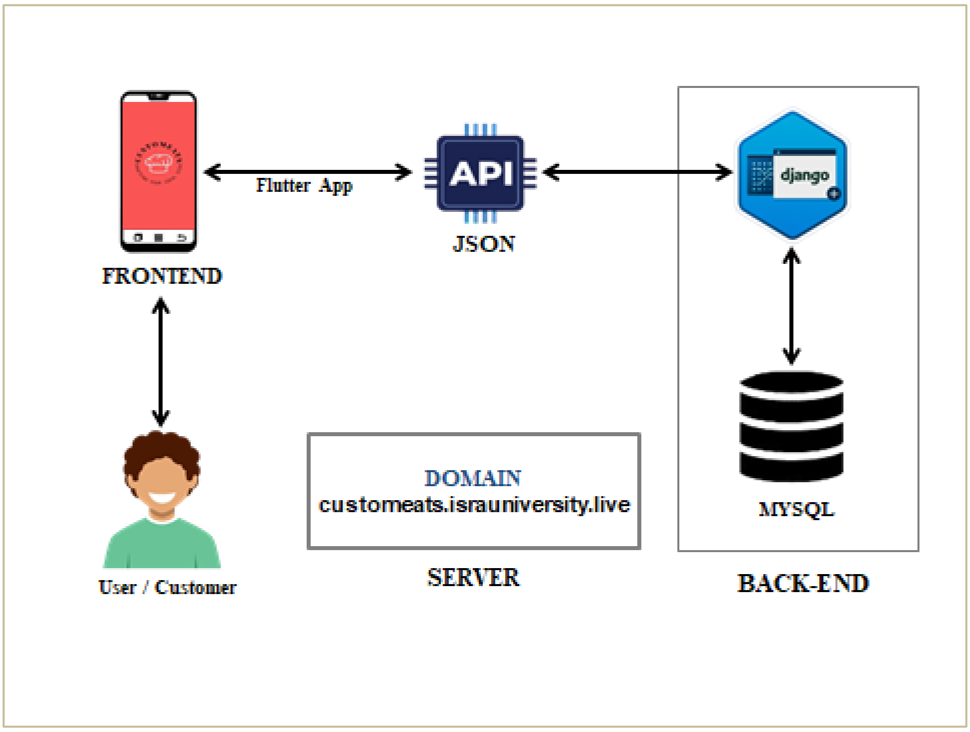


Figure 3.4.2 System Architecture Diagram

3.5 Future Enhancements

In the last and final step of methodology comes the future enhancement, we all know when a software or an application is developed it's not completely correct, neither it can be, so we keep improving it. The future enhancements include, help & support, delivery options (so that Favorites functionality could be used), integration of admin interface, personalized meal suggestions, categories based on religious preferences, allergen and dietary filters and enhanced payment options.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 System Organization

The system architecture for this food customization app integrates both front-end and back-end components to ensure seamless functionality, and to provide a smooth and efficient user experience. Below is a hierarchical diagram representing the key components and their connections.

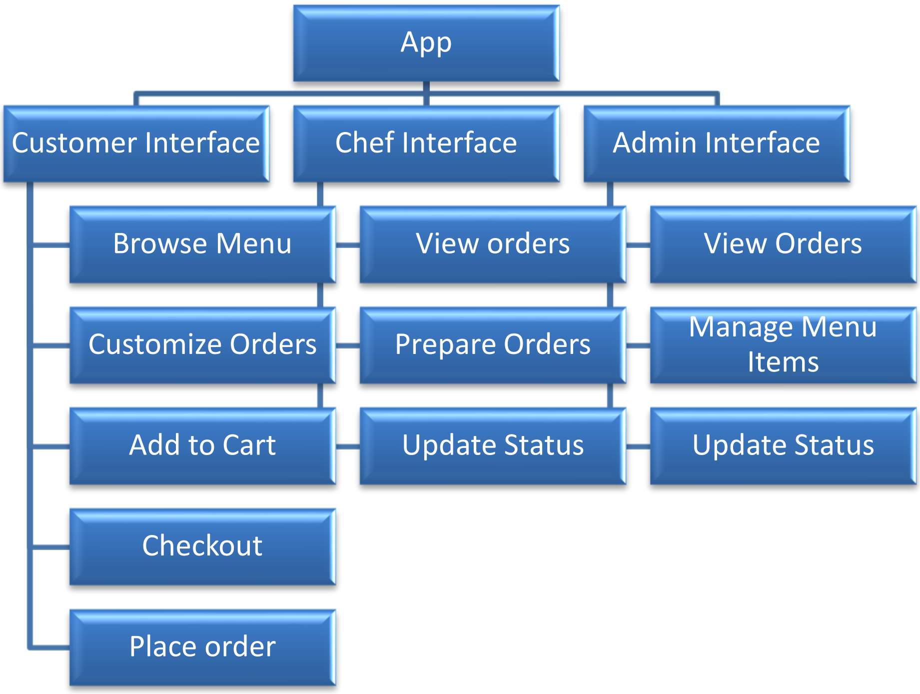


Figure 4.1 System Organization

Let's break-down the system organization to get a better understanding of each component and their functionality.

4.2 System Organization Break-Down

The system organization breakdown provides a detailed overview of the app's process flow, showcasing how each interface Customer, Chef, and Admin interacts to deliver seamless functionality.

4.2.1 Customer Interface

As it can be seen, customer interface handles the ordering of meals by allowing customers to customize a food item according to their preferences. Customers can browse menu, select an item they like, customize the ingredients and spice level as they prefer, either add it to cart or save it to favorites for future orders, in the cart customers can view customizations they made and if needed they can also change them, then precede to payment and complete the order.

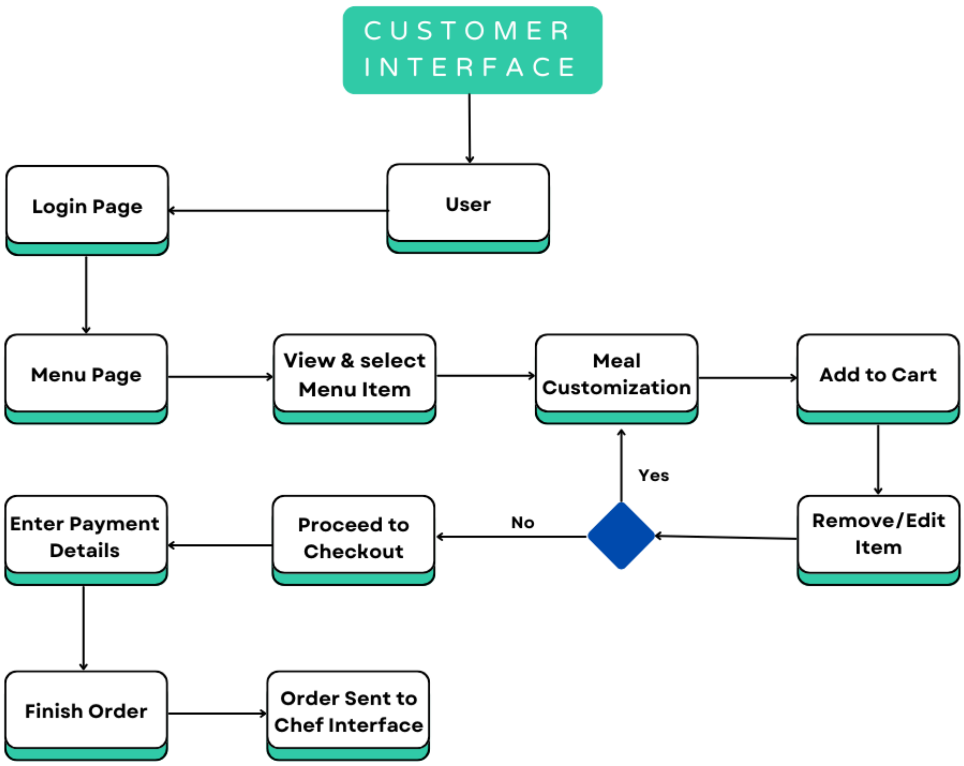


Figure 4.2.1 Customer Interface Process Flow

4.2.2 Chef Interface

Secondly we have chef interface, where the chef has to login first to see incoming orders. The chef can select an order to view the customization details, start preparing an order and update the order status as “in progress”, again update the status as “finish” when the order is ready, and inform the front desk so the order can be delivered/served to customer table. And so on, continue to prepare next orders.

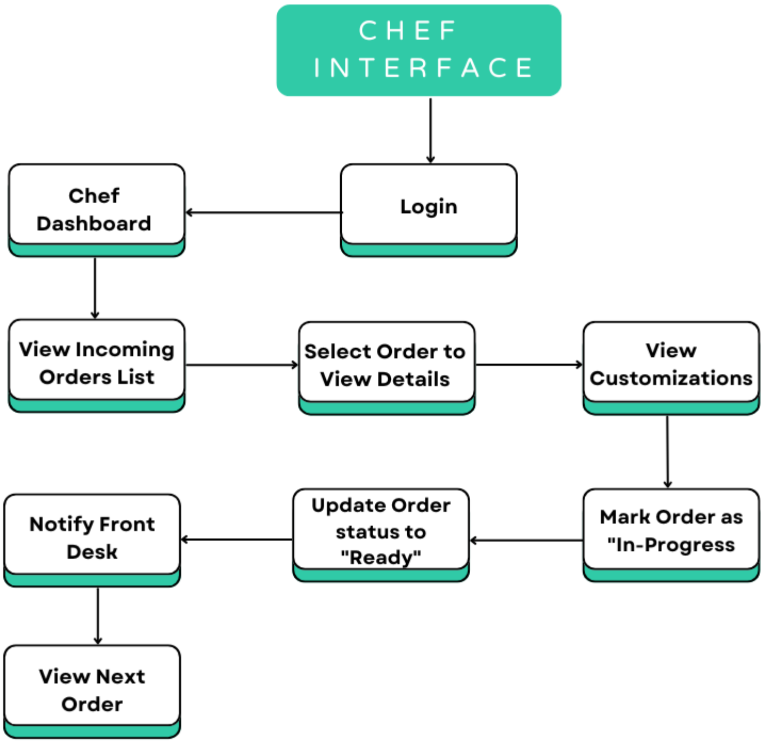


Figure 4.2.2 Chef Interface Process Flow

4.2.3 Admin Interface

Lastly, we have admin interface, where the admin has to login in first to get access to the dashboard. The admin can manage staff (like adding new staff details or removing for the ones who have left), view and manage menu, add or remove items, update prices and availability of items, can view incoming orders, check order status (either pending, in progress, or ready), can also view order details (like time, date, payment method etc.).

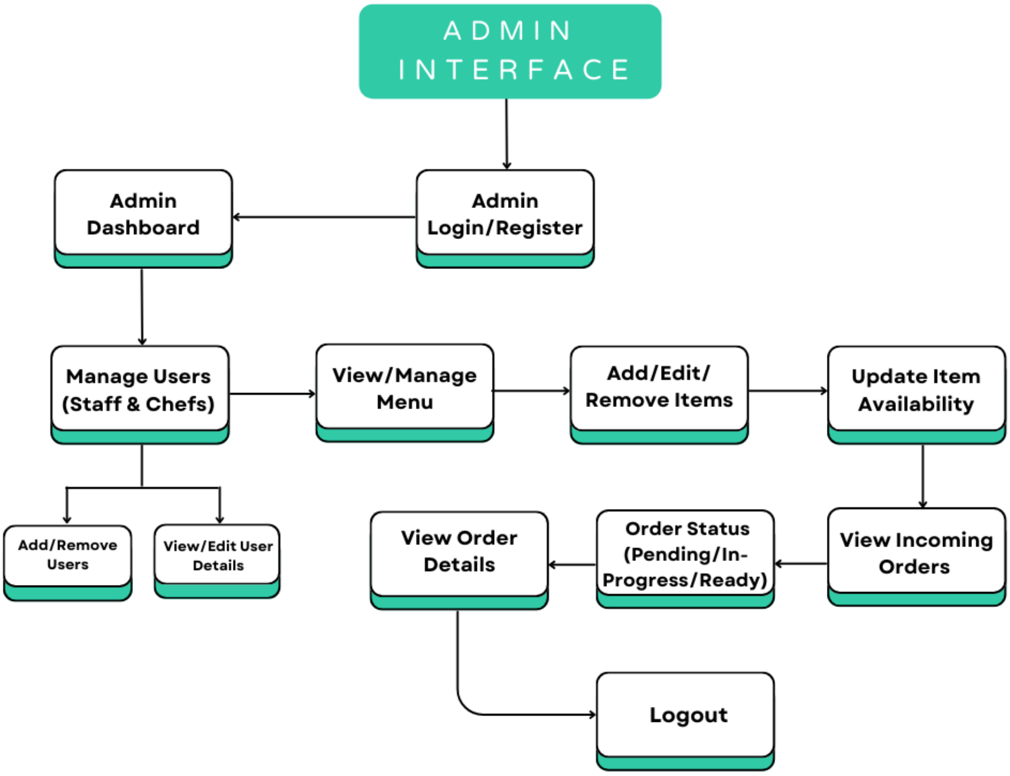


Figure 4.2.3 Admin Interface Process Flow

4.3 App Features

The purpose of CustomEats is to enhance the ease of meal customization and make it more inclusive. From tweaking spice levels, dietary filters, and ingredient choices to order updates, the options are endless with this app. The app also guarantees clarity through its ingredients provided and allows users to save their favorite tweaks to help them in the future.

4.3.1 Login & Logout

A user (which is table for now) has to be logged-in in the device so that when the chef receives the order he/she would know where this order came from. Since this app focuses on a tablet based dine-in restaurant for now, all the devices will already be logged-in by an authorized member. And if any device has to be sent for repairing, the device will be logged-out again by the authorized member.

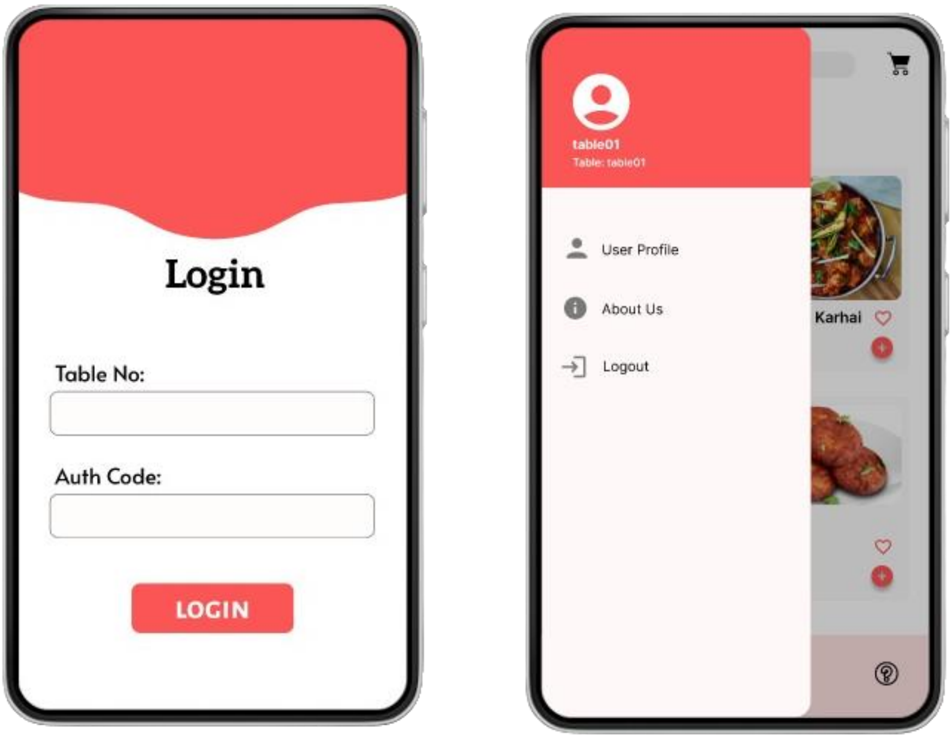


Figure 4.3.1 Login and Logout

4.3.2 Ingredients Selection / Meal Customization

After the customer has selected an item of their choice from menu, for suppose customer has selected Biryani, customer can select protein, rice type, oil type, side dish, and, spice level of their choice. The customer can also can add extra ingredients if they want. As it can be seen, only two fields are marked as required so that those customers who do not want much of a customization can skip it, and when optional ingredients are not selected from the given lists the chef would go with the default ingredients.

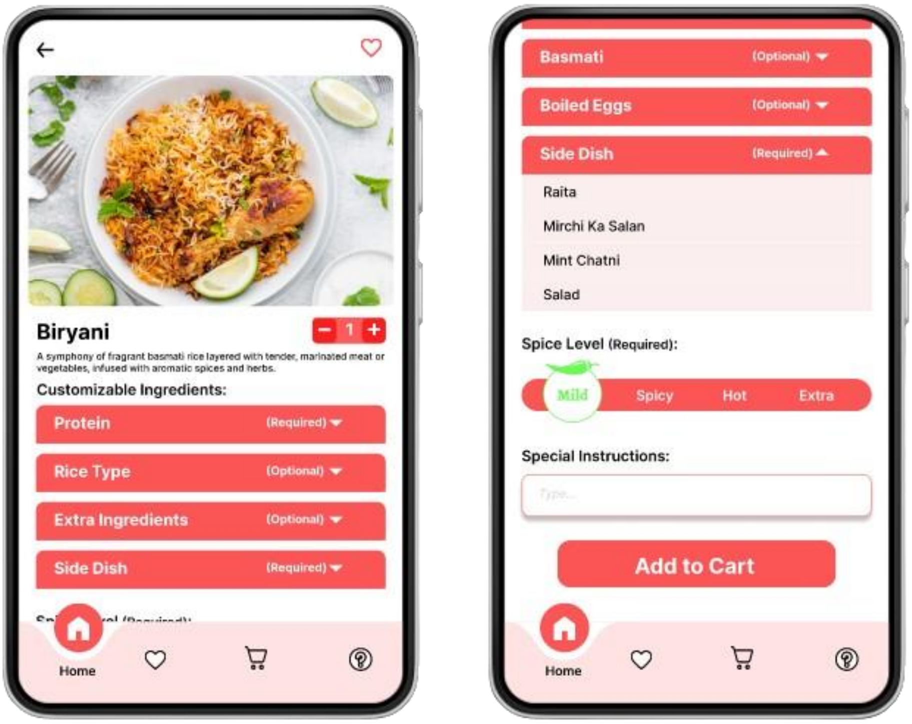


Figure4.3.2 Ingredients Selection

4.3.3 Cart Feature

After customer has added an item to the cart, they can see the customizations they made by clicking on ‘View Details’, and if they want to modify the customizations they made, for suppose, selecting spice level from mild to hot, they can tap on the item and will be navigated back to the ingredients selection, where they can modify the ingredients of the dish.

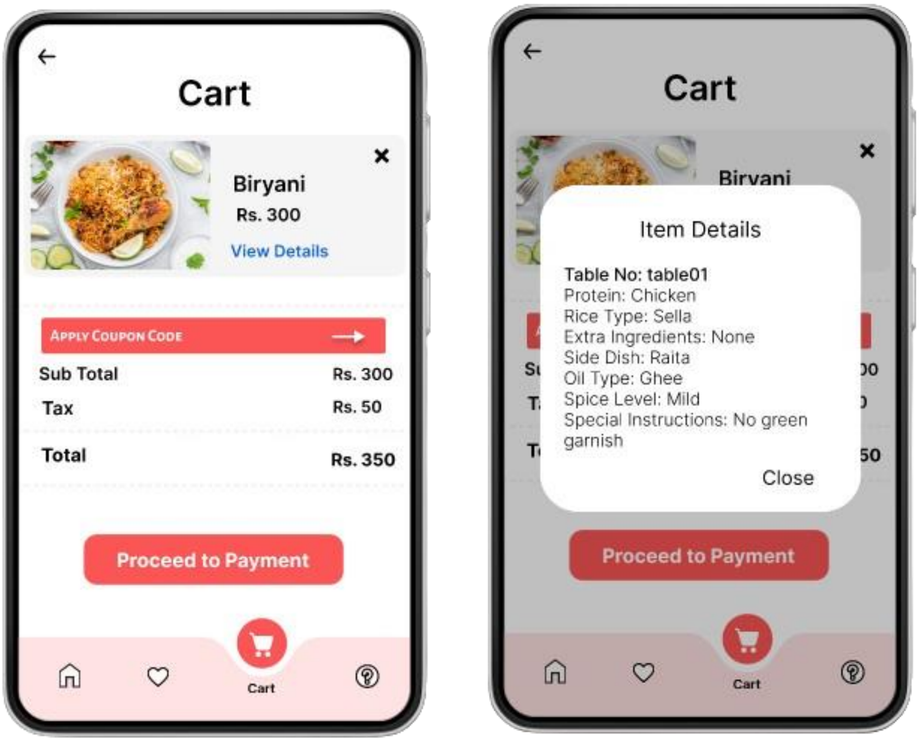


Figure 4.3.3 Cart Feature

4.3.4 Favorite Feature

Favorite feature of this app allows customers to save their customized meals, so customers can smoothly order the same meal in future without having to customize it As the proposed solution is focuses on a dine-in table based restaurant, where customers will have to use tablets available on each table, to order what they want. Because of this, the Favorite feature is totally useless, as customers cannot save their customized meals because there is no use for it. Let's say the same customer visits again, but who knows if the table he sat on last time would be available. But this functionality will be soon useful to customers, when this app will be available for them to install and when delivery options will be added.

Figure 4.3.4 Favorite Feature

4.3.5 Payment Methods

In today's fast pacing world, money is becoming virtual and not everyone carries cash with them, so an app has to have multiple payment options (like credit/debit cards, online payment, money transfer etc.) so that customers do not have to feel frustrated or dissatisfied. This is what we have tried to achieve.

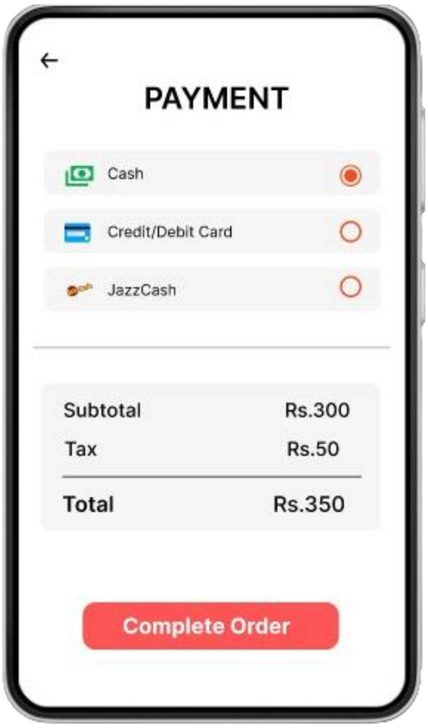


Figure 4.3.5 Payment Methods

4.3.6 Order Status

After the customer is done placing order, the order will be sent and displayed in the kitchen on chef's screen. When the chef will start preparing the dish he/she will click on "Start" button on the KDS (Kitchen Display System) under each order. That time the customer on his/her screen can see that their order is in process. Just like that when that order will be ready the chef will click on "Finish" button and the customer will know that their food is ready and is about to be served.

The green color indicates the current status of the order, like the figure 4.3.6 below shows that the order is in process.

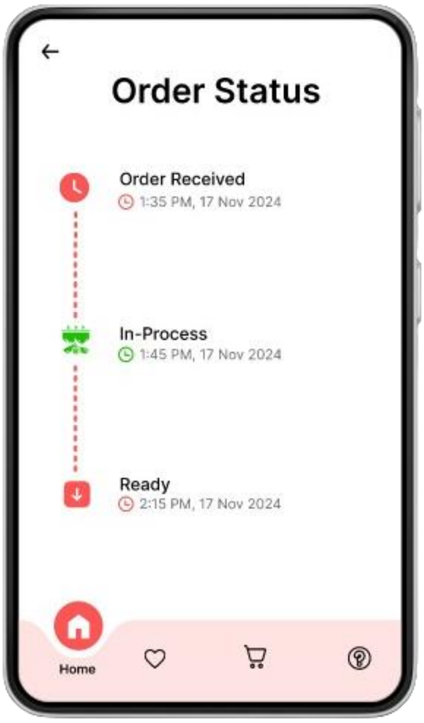


Figure 4.3.6 Order Status

CHAPTER 5

CONCLUSION

5.1 Introduction

The purpose of CustomEats: “Crafting your Ideal Dish”, is to provide customers or users with the best food customization possible, and make them more confident in their food choices. Using CustomEats customers can customize a meal according to their preferences, whether the customer is a vegetarian, picky eater, health conscious, or someone who respects their religious restrictions. This app aims to provide customers with alternative options, in a dish they have wanted to try, but had to skip it because of a single ingredient that could not be replaced.

CustomEats was just an idea, a FYP (Final Year Project), until the survey was conducted, so that we could know more about real-world problems faced by individuals while ordering food or eating at a restaurant. Many customers expressed their frustrations and dissatisfaction with current food apps (already discussed in chapter 2, 2.3) which confirmed the need for an app like CustomEats, which is more dynamic, intuitive, and inclusive when it comes to food customization.

Interpretation of results or we can say how findings contributed in the app’s development. The findings showed how frustrated individuals were with current food apps, how there were not enough vegetarian, and customization options. Some individuals even said that customization should be a basic right for everyone without having to pay extra. CustomEats focuses on providing everything possible to its customers, like vegetarian option in each dish, no extra charges for customization, making sure the food is halal and has alternative options.

CustomEats is limited to providing customization in only desi and desi fusion food items. Based on our previous discussion and research we saw enough apps that provided customization in fast food, like burgers and pizza. The development limitations of CustomEats app include, no delivery options, app not available for customers to install, no reviews or feedback system for customers to give or see, and customization can only be made from given choices though the customer can ask for some ingredients to be more or less. But it’s all part of future enhancements.

If we talk about real-world applications of this project, it helps three categories, restaurants, chefs, and customers. Let’s discuss each in detail, firstly we have restaurants, this app could benefit a business by increasing its sales and by gaining it loyal customers, because when the food is fresh and according to customer preferences and most importantly without any extra charges, who will not come again to such place.

Secondly, chefs, who most of the time struggle because of incorrect food orders or because of lack of direct communication with customer, through this app chef would know the exact customization customer asked for and if anything goes wrong chef will not be the one to blame.

Lastly we have customers, who are the basic priority of this app, this app benefits each type of customers because customers can customize their food however they want, either healthy, dietary, vegetarian, non-vegetarian etc.

5.2 Future Enhancement

To make CustomEats more efficient and usable for users, the following changes/features will be added,

i. **Reviews & Feedback**
   Through user reviews and feedback other customers can see if the customers earlier received food with correct customization and if the food was delicious or not.

ii. **Delivery Options**
   No matter how great of the customization an app provides, but until it does not have delivery options it will not satisfy its customers, since not everyone likes to eat at public places and if there would be delivery options it would save customers time and energy.

iii. **Filters**
   Individuals like those who work late shifts or those who value their time more than money prefer features like search and filters to quickly get what they want.

iv. **Nutrition Based Recommendations**
   More people these days focus on eating food that is both healthy and less greasy and what could be better than having both customization and nutrition in one single app.

v. **Allergen Alerts**
   A dish could contain some ingredients that the restaurant cannot tell about, to keep the recipe a secret, but it’s a restaurants responsibility to make sure their customers’ safety by warning them about the ingredients that possibility could cause allergies.

vi. Food Categories

Introducing categories based on religions would help those individuals who struggle to find halal food or the food the best fits their religious restrictions.

To cut a long story short, CustomEats was able to fill a gaping hole in the food sector with a meal customization experience suitable to the dietary needs and preferences of customers, offering a range of choices from vegetarian to halal options, to minimizing excessive charges on customizations and putting customers first. Even though the features like delivery and feedback are not included, the planned future changes look to fine tune CustomEats as the holistic, user centric dining solution to meet different wants and needs.

REFERENCES

[1]. **Jennifer Smith Tapp**. (2023, September 15). *The Evolution of Customizable Menus at Fast-Food Chains*. *Eyes4Research Reports*. Retrieved from https://www.eyes4research.com/the-evolution-of-customizable-menus-at-fast-food-chains/

[2]. **Kempinski, A. J., & De Vries, D.** (2019). *A Critical Review of Food Customization Impacts on Consumer Behavior*. *Critical Reviews in Food Science and Nutrition*, 59(18), 2940-2950. Retrieved from https://www.tandfonline.com/doi/abs/10.1080/10408398.2018.1559796

[3]. **Tastewise**. (2024, August 22). *The Rise of Food Personalization and Its Impact on Menu Planning*. *Tastewise Blog*. Retrieved from https://tastewise.io/blog/food-personalization

[4]. **Nagle, P.** (2024, June 12). *At-Home Experiences & Customization Emerging as Innovation Trend in Food Service*. *Food Business News*. Retrieved from https://www.foodbusinessnews.net/articles/25975-opinion-at-home-experiences-customization-emerging-as-innovation-trend

[5]. **Research Publish Journals**. (2022, February 5). *Consumers' Perception and Satisfaction with Food Customization Options*. Retrieved from https://www.researchpublish.com/upload/book/Consumers%E2%80%99%20Perception-05022022-1.pdf

[6]. **Typeset**. (n.d.). *How Does the Customization of Meals Impact Customer Preferences?* Retrieved from https://typeset.io/questions/how-does-the-customization-of-meals-impact-customer-5e4bp4o9tp

[7]. **Dribbble**. (2024). *Recipe or Food Order Kitchen Display System*. Retrieved from https://dribbble.com/shots/24298385-Recipe-or-Food-Order-Kitchen-Display-System-DailyUI-40

[8]. **Ashraff, I.** (n.d.). *McDonald's Online Ordering System*. *GitHub Repository*. Retrieved from https://github.com/iamashraff/McDonalds-Online-Ordering-System

APPENDIX A

This code below provides functions to:

- API fetch functions for each customization option (protein, rice type, extra ingredients, side dish, oil type, and spice level).
- The sendCustomizationDetails function to send selected customization data to the server after a user finalizes their choices.
- Save the selected customizations in SharedPreferences (to represent adding to a cart).

```
import 'package:http/http.dart' as http;

import 'dart:convert';

import 'package:shared_preferences/shared_preferences.dart';

import 'package:flutter/material.dart';

// Function to fetch options from each API endpoint

Future<List<String>> fetchOptions(String url) async {

  final response = await http.get(Uri.parse(url));

  if (response.statusCode == 200) {

    List<dynamic> data = json.decode(response.body);

    return data.map<String>((item) => item['name'].toString()).toList();

  } else {

    throw Exception('Failed to load options');

  }

}

// Function to fetch all ingredient options and set their lists

Future<void> fetchData() async {  proteinOptions = await
fetchOptions('http://customeats.israuniversity.live:8000/api/protein/');

  riceTypeOptions = await
fetchOptions('http://customeats.israuniversity.live:8000/api/ricetype/');
```

```
  extraIngredientsOptions = await
fetchOptions('http://customeats.israuniversity.live:8000/api/extraingredients/');

  sideDishOptions = await
fetchOptions('http://customeats.israuniversity.live:8000/api/sidedish/');

  oilTypeOptions = await
fetchOptions('http://customeats.israuniversity.live:8000/api/oiltype/');

  spiceLevelOptions = await
fetchOptions('http://customeats.israuniversity.live:8000/api/spicelevel/');

}

// Function to send customization details to the server

void sendCustomizationDetails(BuildContext context, Map<String, dynamic> item) async {

  final url = Uri.parse('http://customeats.israuniversity.live:8000/api/receive-
customizations/');

  Map<String, dynamic> customizationData = {

    'protein': item['protein'] ?? 'N/A',

    'riceType': item['riceType'] ?? 'N/A',

    'extraIngredients': item['extraIngredients'] ?? 'N/A',

    'sideDish': item['sideDish'] ?? 'N/A',

    'oilType': item['oilType'] ?? 'N/A',

    'spiceLevel': item['spiceLevel'] ?? 'N/A',

    'specialInstructions': item['specialInstructions'] ?? 'N/A',

    'tableNo': item['tableNo'] ?? 'N/A',

  };

  try {

    final response = await http.post(

      url,

      headers: {'Content-Type': 'application/json'},

      body: jsonEncode(customizationData),
```

```
    );
    if (response.statusCode == 200) {
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text("Customizations sent to server successfully!")),
      );
    } else {
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text("Failed to send customizations to server")),
      );
    }
  } catch (e) {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text("Error occurred: $e")),
    );
  }
}
```

APPENDIX B

- This function _showItemDetails creates a dialog box showing the customization details saved for each item when 'View Details' is clicked.

```
void _showItemDetails(Map<String, dynamic> item) {
 showDialog(
   context: context,
   builder: (BuildContext context) {
     return AlertDialog(
       title: Text('Item Details'),
       content: SingleChildScrollView(
         child: ListBody(
           children: <Widget>[
             Text('Protein: ${item['protein'] ?? 'N/A'}'),
             Text('Rice Type: ${item['riceType'] ?? 'N/A'}'),
             Text('Extra Ingredients: ${item['extraIngredients'] ?? 'N/A'}'),
             Text('Side Dish: ${item['sideDish'] ?? 'N/A'}'),
             Text('Oil Type: ${item['oilType'] ?? 'N/A'}'),
             Text('Spice Level: ${item['spiceLevel'] ?? 'N/A'}'),
             Text('Special Instructions: ${item['specialInstructions'] ?? 'N/A'}'),
             Text('Table No: ${item['tableNo'] ?? 'N/A'}'), // Displaying table number
           ],
         ),
       ),
       actions: <Widget>[
```

```
          TextButton(
            child: Text('Close', style: TextStyle(color: Colors.black)),
            onPressed: () {
              Navigator.of(context).pop();
            },
          ),
        ],
      );
    },
  );
}
```

APPENDIX C

- This function checks if the selected item with its specific customization details (like protein, riceType, etc.) already exists in favorites. If not, it saves the item to both the local favorites list and SharedPreferences.

```
Future<void> _addToFavorites(Map<String, dynamic> item) async {

 SharedPreferences prefs = await SharedPreferences.getInstance();

 List<String>? favoritesList = prefs.getStringList('favorites') ?? [];



 // Check if the item is already in favorites to prevent duplicates

 bool itemExists = favorites.any((favorite) =>

     favorite['itemname'] == item['itemname'] &&

     favorite['protein'] == item['protein'] &&

     favorite['riceType'] == item['riceType']); // Extend with other criteria as needed

 if (!itemExists) {

   // Add item to local list

   setState(() {

     favorites.add(item);

   });

   // Save updated favorites list to SharedPreferences

   favoritesList.add(json.encode(item));

   await prefs.setStringList('favorites', favoritesList);

   ScaffoldMessenger.of(context).showSnackBar(

     SnackBar(content: Text('Item added to favorites')),

   );
```

```
  } else {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text('Item is already in favorites')),
    );
  }
}
```